BrainNet: A Systematic Benchmark for Functional Brain Network Generation and Analysis

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Abstract

Functional interactions between brain regions can be viewed as a network, enabling neuroscientists to investigate brain function through network science. We introduce BrainNet, a systematic benchmark for constructing ROI-based functional brain networks from resting-state fMRI. Thirty-three pipelines are evaluated, spanning node parcellations; connectivity definitions (pairwise, higher-order, and learned edges); sparsification/thresholding strategies; and static versus dynamic modeling, all under a unified training–validation protocol with a fixed graph neural network classifier. Results show substantial variability across pipelines; notably, sparsification choices can invert method rankings, underscoring the importance of density control. Nonetheless, a compact set of configurations consistently meets reliability and performance criteria across ADNI, HCP S1200, and ADHD-200. We introduce practical guidelines for reproducible connectome generation, open resources, and implications for downstream graph-based machine learning and clinical research, with documentation available at `https://brainnet-fmri-codedocs.readthedocs.io` and the open-source package at `https://pypi.org/project/brainnet-graph`.

Keywords: fMRI, brain graph, graph neural network, graph construction.

1 Introduction

The human brain is a complex, interconnected system of regions, and understanding its function requires methods that can capture inter-regional interactions. Over the past decade, neuroscientists have increasingly modeled brain activity as a network (or graph) of functionally interacting regions, embracing tools from network science to study brain function and dysfunction [6, 13]. In this framework, different parts

of the brain are represented as nodes in a graph, and statistical relationships between their activity time series serve as edges. This graph-theoretic approach—often referred to as functional connectomics—has yielded valuable insights into how brain network organization underlies cognition, behavior, and neuropathology [34, 71]. In particular, noninvasive techniques like resting-state functional MRI (rs-fMRI) have become pivotal: rs-fMRI captures spontaneous brain dynamics without a task, enabling researchers to derive whole-brain functional connectivity patterns that reveal the brain's intrinsic network architecture [87].

A critical step in functional connectomics is defining the brain's regions of interest (ROIs) and constructing a graph from fMRI data. Typically, the continuous brain volume is partitioned into ROIs according to an anatomical or functional atlas, or via data-driven parcellation methods [5]. Each ROI serves as a node in the resulting brain graph, and its node-level fMRI signal is obtained by aggregating the BOLD time series of voxels within that region. An edge between two nodes is then defined by a measure of functional connectivity between their respective time series—for example, the Pearson correlation of the BOLD signals of two ROIs is a common choice for edge weight [87]. This abstraction of the brain into a network of ROIs dramatically reduces data dimensionality and enables the application of graph theory metrics (e.g., clustering coefficient, path length) to characterize brain organization. However, it is important to note that ROI definition is not uniquely determined: the brain can be parcellated at different granularities (from tens to hundreds of regions), and ROIs can be defined as non-overlapping parcels or as overlapping networks, with no clear consensus on the "best" parcellation scheme [5]. These choices in node definition can alone induce substantial differences in the derived networks. Similarly, functional edges can be quantified in various ways (Pearson correlation, partial correlation, mutual information, coherence, etc.), and researchers may apply different normalization, thresholding, or sparsification techniques when constructing graphs. Each of these decisions can alter the topology and weights of the resulting brain network, highlighting that functional connectome construction is a highly nontrivial process with many degrees of freedom [72].

Crucially, the choices made in building functional brain networks have a major impact on subsequent analyses and findings. Recent studies have demonstrated that different fMRI network construction pipelines—varying in ROI definitions, connectivity metrics, and preprocessing steps—can yield markedly different networks from the same raw data, sometimes leading to contradictory conclusions about brain function [22, 61]. For example, [61] systematically evaluated hundreds of pipeline variations (combining multiple parcellations, connectivity measures, and other processing options) and found vast variability in outcomes: the majority of pipelines failed basic reliability criteria (e.g., producing networks that were not reproducible across test–retest scans), implying that many common analysis choices can produce spurious or misleading results. Similarly, benchmarking work by [22] showed that the selection of ROI parcellation and connectivity measure significantly affects predictive modeling performance across multiple clinical datasets, even when using identical machine learning algorithms. In particular, certain pipelines consistently outperformed others in classification tasks (e.g. discriminating patients from healthy controls), underscoring that graph construction is not just a minor preprocessing detail but rather a determinant of downstream success. These findings, along with other comparative studies, underscore the broader point that functional connectomics results are highly sensitive to methodological choices [61, 72]. A poor choice of network construction method can distort network topology, obscure true brain-behavior relationships, and ultimately hinder the discovery of reliable neuroimaging biomarkers. Conversely, identifying robust graph construction practices is essential to ensure that network analyses reflect genuine neurobiological signals rather than methodological artifacts.

In parallel with these developments in neuroscience, the machine learning community has increasingly turned to graph neural networks (GNNs) and other deep learning approaches to analyze brain connectivity data. GNNs are powerful models that operate on graph-structured inputs, learning representations of nodes, edges, or entire graphs for tasks such as classification and regression. This paradigm has opened new avenues for disorder classification, outcome prediction, and other data-driven analyses of brain networks [36, 57]. For instance, recent studies have applied GNNs to rs-fMRI–derived brain graphs to successfully distinguish between clinical populations (e.g., autism or schizophrenia patients vs. controls) and to identify salient network features associated with disease [57]. Importantly, the performance and interpretability of these graph-based models depend directly on the quality of the input graphs. If the functional brain network fed into a GNN is noisy or constructed suboptimally (for example, missing important connections or including spurious ones), the model may learn incorrect patterns or fail to generalize. In other words, even the most sophisticated GNN will be bottlenecked by the fidelity of the underlying brain graph. This links back to the need for reliable network construction: improving the graph itself can translate to better machine learning outcomes, more sensitive disease detection, and

more meaningful neuroscientific interpretations of learned features. As GNNs and other advanced algorithms become more common in connectomics, establishing sound graph generation practices becomes ever more critical for maximizing their impact.

Despite the clear importance of how functional brain networks are generated, the field currently lacks standardized guidelines or consensus on the optimal construction pipeline. There is tremendous methodological heterogeneity across studies: labs use different atlases or parcellation techniques, apply different connectivity definitions (*e.g.* correlation vs. information-theoretic measures), and even diverge on fundamental preprocessing steps like filtering or global signal regression. This lack of standardization has led to inconsistent findings and difficulties in reproducibility when results are compared across datasets or research groups [22, 72]. For both neuroscience researchers and machine learning practitioners entering the neuroimaging domain, this fragmentation poses a significant challenge: it is often unclear which of the many available graph construction methods will yield reliable and biologically meaningful results for a given application. Benchmarking efforts are needed to bring clarity and objectivity to this problem by evaluating how different network construction choices affect outcomes across diverse conditions. While some prior works have compared specific aspects of connectome construction (for example, studying how thresholding affects particular graph metrics, or how one parcellation compares to another in a limited setting), there has not yet been a comprehensive, end-to-end evaluation spanning the full pipeline space across multiple datasets and tasks.

To address this critical gap, we present *BrainNet*: a systematic benchmark for functional brain network generation and analysis. To our knowledge, BrainNet is the first effort to rigorously evaluate and compare a broad range of graph construction pipelines under a unified framework, assessing their impact on various downstream tasks and across multiple neuroimaging datasets. By doing so, we aim to establish an empirical basis for recommending best practices in functional connectomics and to provide the community with open tools and resources for reproducible network analysis. In summary, our contributions are:

- Taxonomy of graph construction methods: We propose a new taxonomy to categorize and describe existing methods for functional brain network construction. This taxonomy clarifies the landscape of choices (in ROI definition, connectivity measures, thresholding/sparsification strategies, etc.) and highlights conceptual differences and commonalities between approaches.
- Comprehensive literature review: We provide an extensive review of prior work in functional connectome construction and analysis. We summarize the evidence on how various methodological choices affect network properties, reliability, and validity, drawing on findings from recent studies (including [61] and others) to motivate the need for systematic benchmarking.
- Standardized benchmark and open resources: We introduce a benchmarking framework that evaluates a wide array of network construction pipelines on multiple datasets and tasks. BrainNet includes evaluations on diverse classification problems (e.g., disease vs. control), assessments of graph theoretical properties (e.g., community structure, efficiency), and interpretability analyses. All code, datasets, and results are made publicly available to serve as a standardized testbed for future method development in functional connectomics.
- Practical guidance for method selection: Based on our systematic experiments, we derive actionable guidelines for researchers. We identify which graph construction techniques tend to perform best for certain scenarios (e.g., which connectivity metric yields the most reliable networks for clinical classification, or which parcellation scale balances detail and robustness). This guidance is intended to inform both neuroscientists building connectomes and ML practitioners applying GNNs or other models, helping them choose appropriate network generation pipelines for their specific goals.

By establishing this benchmark, we hope to accelerate progress toward more reproducible, comparable, and effective functional brain network analyses, ultimately bridging the gap between methodological experimentation and practical recommendations for the community.

2 Background and Definition

2.1 Background

A Brief History of Functional Brain Networks

The concept of the brain as a network of interconnected regions has evolved over more than a century, with its origins rooted in early neuroscience efforts to map neuronal structure and function. In the late 19th and early 20th centuries, foundational work by Camillo Golgi and Santiago Ramón y Cajal utilized staining techniques to reveal the intricate connections between neurons. Golgi's silver chromate method enabled visualization of neuronal networks, while Cajal's advocacy for the neuron doctrine established neurons as the brain's fundamental functional units, interconnected via synapses [33]. These studies primarily focused on structural connectivity, setting the stage for later explorations of functional interactions.

The modern understanding of functional brain networks emerged in the 1990s with the advent of functional magnetic resonance imaging (fMRI). A landmark contribution came in 1994 when Karl Friston formalized the concept of "functional connectivity" as the temporal correlation between time series of distinct brain regions [35]. This definition provided a framework for investigating dynamic communication across the brain, shifting focus from static anatomy to functional relationships.

In 1995, Bharat Biswal and colleagues demonstrated that even in the absence of tasks, resting brain activity exhibited correlated patterns, such as between motor cortices [10]. This finding challenged the prevailing assumption that resting brain activity was unstructured noise, instead suggesting the presence of intrinsic functional networks. Biswal's work is widely recognized as the inception of resting-state fMRI, a field that has since grown to dominate functional connectivity research.

A significant milestone occurred in 2001 when Marcus Raichle and colleagues identified the default mode network (DMN), a set of regions—including the medial prefrontal cortex, posterior cingulate cortex, and lateral parietal cortices—that are more active during rest than during task performance [77]. The DMN is implicated in self-referential thought, memory retrieval, and prospection, highlighting the brain's intrinsic activity during idle states. In 2003, Michael Greicius and colleagues further characterized the DMN using resting-state fMRI, confirming its functional coherence and expanding its implications for network neuroscience [41].

Since the early 2000s, the field has identified additional large-scale networks, such as executive control and salience networks, leveraging advances in graph theory and computational methods to map and quantify connectivity at multiple scales. The study of functional brain networks has become a cornerstone of neuroscience, offering insights into brain organization, cognition, behavior, and the pathophysiology of neurological and psychiatric disorders [33]. This historical progression reflects a paradigm shift from viewing the brain as a collection of isolated regions to recognizing it as a complex, interconnected system, with ongoing research continuing to refine our understanding of its dynamic properties.

Table 1: Key Milestones in the History of Functional Brain Networks

Year	Milestone	Key Contribution	Reference
1890s	Golgi and Cajal's work on neuron staining	Foundation for neuronal connectivity	[33]
1994	Friston defines functional connectivity	Framework for brain communication	[35]
1995	Biswal et al.'s resting-state fMRI study	Evidence of intrinsic networks	[10]
2001	Raichle et al. propose the DMN	Identification of resting-state network	[77]
2003	Greicius et al. characterize the DMN	Coherence of DMN established	[41]

A History of ROI-Based Functional Connectivity and Functional Brain Graphs

Building on the foundational work in functional connectivity, the region of interest (ROI)-based approach has emerged as a powerful method to study brain networks by focusing on specific, predefined areas of the brain. This approach traces its roots to the broader application of fMRI in the 1990s, where early studies began to segment the brain into discrete regions to analyze their interactions more systematically. The shift toward ROI-based analysis was driven by the need to reduce the complexity of whole-brain voxel-wise analyses and to align functional data with anatomical or functional hypotheses about specific brain regions.

The formalization of ROI-based functional connectivity gained traction in the early 2000s alongside advancements in resting-state fMRI. For instance, following the identification of the default mode network (DMN) by [77], researches like [41] employed ROI-based methods to investigate the DMN's connectivity patterns, using seed regions such as the posterior cingulate cortex to correlate activity with other brain areas. This seed-based approach, where a single ROI's time series is correlated with all other voxels or ROIs, became a cornerstone of early ROI-based studies, offering a straightforward way to map network interactions.

Concurrently, the integration of graph theory into neuroscience, popularized in the late 1990s and

early 2000s by researches like [13], catalyzed the development of functional brain graphs. By representing ROIs as nodes and their functional connectivity as edges, investigators could apply graph-theoretic metrics—such as clustering coefficient, path length, and modularity—to characterize the brain's topological organization. A pivotal study [13] demonstrated that functional brain networks exhibit a small-world architecture, balancing local specialization and global integration, a finding that has since been replicated across numerous ROI-based analyses.

The evolution of ROI-based methods accelerated with the availability of standardized brain atlases, such as the Automated Anatomical Labeling (AAL) atlas, which provided predefined anatomical ROIs for consistent analysis across studies [21]. Around the same time, data-driven approaches like independent component analysis (ICA) emerged as alternatives to anatomically defined ROIs, allowing researchers to identify functionally coherent regions without prior assumptions [8]. These developments marked a significant shift toward flexibility and precision in ROI selection, enabling more nuanced investigations of functional connectivity.

By the mid-2000s, ROI-based functional connectivity and functional brain graphs had become central to studying brain dynamics across the lifespan and in disease states. For example, [31] used ROI-based graphs to map the development of functional networks from childhood to adulthood, while [42] highlighted altered DMN connectivity in Alzheimer's disease. These studies underscored the utility of ROI-based approaches in both basic and clinical neuroscience, paving the way for their widespread adoption.

Today, ROI-based functional connectivity and functional brain graphs are integral to network neuroscience, supported by sophisticated computational tools and multimodal imaging techniques. Ongoing research continues to refine these methods, addressing challenges such as ROI definition, dynamic connectivity, and integration with structural data, to further elucidate the brain's complex functional architecture.

Despite significant advancements, the field of functional brain graph analysis faces ongoing challenges and promising future directions. A key issue is the definition of regions of interest (ROIs), where the choice can heavily influence outcomes, yet no consensus exists on the best approach, prompting exploration into data-driven methods like independent component analysis (ICA) for more objective ROI delineation, as highlighted by Craddock (2012) and Beckmann (2005). Additionally, while most research has centered on static connectivity, the brain's functional connectivity is inherently dynamic, shifting over time, which has spurred active investigation into methods for studying and interpreting these temporal changes, as noted by Chang (2010) and Hutchison (2013). These challenges underscore the need for continued innovation to refine our understanding of the brain's complex, evolving network dynamics [8, 17, 21, 48].

In conclusion, research on ROI-based functional connectivity and functional brain graphs has significantly enhanced our understanding of the brain's functional architecture. Continued advancements in methodology and the integration of findings from various disciplines will further illuminate the complex dynamics of brain networks.

2.2 Definition

Throughout this paper, we adopt the convention of using bold uppercase characters (e.g., $\mathbf{A}$) to represent matrices and bold lowercase characters (e.g., $\mathbf{v}$) to represent vectors. Unless otherwise specified, the notations used are summarized in Table 2. Below, we introduce the minimal set of definitions necessary to understand the concepts presented in this work.

Definition 1 (Graph): A graph G is a mathematical structure that models pairwise relationships between objects. It is defined as $G = (V, E)$, where V denotes the set of nodes (or vertices) and E denotes the set of edges (or connections) between pairs of nodes. In the context of brain networks, G represents regions of interest (ROIs) as nodes and their interactions as edges.

- *Node:* A node $v \in V$ represents an individual entity, such as a brain region.
- *Edge:* An edge $e_{ij} \in E$ represents a connection between two nodes v_i and v_j in the graph.

Definition 2 (Connectivity): Connectivity refers to the degree to which nodes in a graph are interconnected. It is commonly quantified using the adjacency matrix $\mathbf{A}$, where the element A_{ij} indicates the presence or strength of an edge between nodes v_i and v_j.

- *Functional Connectivity:* It is defined as the statistical measure of temporal dependencies between the neuronal activation patterns of anatomically separate brain regions, capturing how these areas interact and exchange information. It is typically represented by an $n \times n$ connectivity matrix, where n is the number of regions of interest (ROI).

Table 2: Commonly Used Notations in Functional Connectivity Analysis

Notation	Description
x_i	The value of the time-series x at a specific time point i.
y_i	The value of the time-series y at a specific time point i.
$\bar{x}$	The mean (average) value of the time-series x.
$\bar{y}$	The mean (average) value of the time-series y.
t	The number of time points in the time-series, used as the upper limit in summations and as a divisor in the formulas for means, covariance, and variance.
n	The number of regions of interest (ROIs) in the brain, which determines the dimensions of the $n \times n$ connectivity matrix derived from pairwise correlations.
r_{xy}	The correlation coefficient, which quantifies the relationship between two time-series, x and y, from two regions of interest (ROIs) in the brain.
$\text{cov}(x, y)$	The covariance between the time-series x and y, measuring how the two variables vary together.
$\text{var}(x)$	The variance of the time-series x, representing the spread of its values around the mean.
$\text{var}(y)$	The variance of the time-series y, representing the spread of its values around the mean.
G	The graph representing the network.
V	The set of vertices (nodes) in the graph.
E	The set of edges in the graph.
v	A vertex in the graph.
e	An edge in the graph.
e_{ij}	An edge connecting vertices i and j.
A	The adjacency matrix of the graph.

Definition 3 (Functional Brain Graph): A functional brain graph is a specialized graph $G = (V, E)$ where the nodes $v \in V$ correspond to brain regions (ROIs), and the edges $e_{ij} \in E$ represent functional connectivities between these regions.

3 Functional Brain Graph Construction

The construction of functional brain graphs from functional magnetic resonance imaging (fMRI) data is a pivotal process in network neuroscience, enabling the transformation of raw neuroimaging data into a structured representation amenable to graph theoretical analysis. This section delineates the fundamental steps involved, underscoring the significance of the graph construction phase—specifically the definition of nodes, edges, and the application of filtering schemes—as these elements profoundly influence the network's topology and the biological insights derived therefrom. Figure 1

3.1 Procedure

The workflow for constructing functional brain graphs typically encompasses the following steps:

1. **Defining Regions of Interest (ROIs)**: ROIs, which constitute the nodes of the graph, are delineated using anatomical landmarks, functional parcellations, or data-driven approaches such as independent component analysis (ICA) [21]. The choice of parcellation scheme is critical, as it impacts the network's structure, inter-subject variability, and interpretability of functional connectivity [87].

2. **Extracting Time Series**: For each ROI, the average blood-oxygen-level-dependent (BOLD) time series across all voxels within the region is computed, encapsulating the temporal dynamics of neural activity in that area.

3. **Computing Connectivity Measures**: Functional connectivity between pairs of ROIs is commonly quantified using the Pearson correlation coefficient of their time series. Alternative metrics,

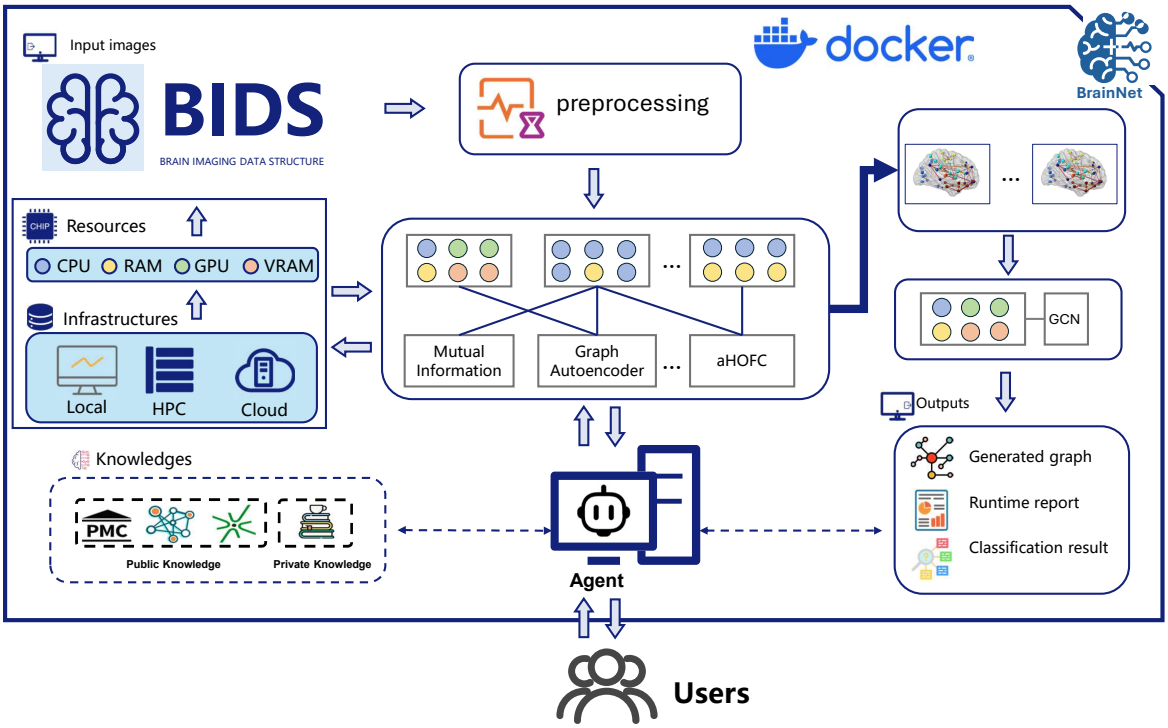

Figure 1: Dockerized end-to-end system for functional brain-network analysis. The entire workflow is packaged in Docker for easy reproduction. Source data are BIDS images. Preprocessing yields ROI-level BOLD time series; a graph-construction module (e.g., Mutual Information, Graph Autoencoder, aHOFC) produces connectivity matrices and surface visualizations; a GCN performs training/inference to return the generated graph and a classification result. Because different methods/stages have distinct compute demands, the system allocates CPU/RAM/GPU/VRAM across local/HPC/cloud infrastructures. A RAG-enabled agent accesses public/private knowledge, orchestrates runs, aggregates logs, and returns runtime reports and explanations to users.

such as partial correlation, mutual information, or coherence, may be employed to capture distinct facets of connectivity, depending on the research objectives [17].

4. **Constructing the Graph**: Each ROI is represented as a node, with connectivity measures defining the edges. This step is central to the process, as the method of edge definition—whether edges are weighted or binarized, and the thresholding approach applied—significantly shapes the graph's properties and downstream analyses. Common thresholding strategies include absolute thresholding (retaining edges above a fixed value) and proportional thresholding (retaining a fixed percentage of the strongest connections).

5. **Graph Analysis**: Post-construction, graph theory metrics such as degree, clustering coefficient, path length, and modularity are calculated to characterize the network's topology. These metrics elucidate properties like small-worldness and modularity, which are essential for understanding brain organization [13]. However, this survey prioritizes the graph construction process over analysis.

3.2 Node Definition

Node choices in fMRI graphs range from voxel/vertex-level to macroscopic regions of interest (ROIs). While voxelwise nodes maximize spatial resolution, ROI nodes dominate practice because they improve interpretability, reduce noise via averaging, and make between-subject analyses tractable [4, 28]. We therefore focus on ROI definitions via *parcellation*.

Parcellations can be (i) **anatomical**, using gyral/cytoarchitectonic landmarks (e.g., AAL, Desikan–Killiany, Lausanne) [14, 25, 90]; (ii) **functional/connectivity-driven**, grouping voxels with similar intrinsic coupling (e.g., Yeo 7/17 networks; Schaefer multi-scale) [80, 98]; (iii) **multimodal**, integrating myelin/architecture/connectivity (e.g., HCP–MMP1.0; Brainnetome) [32, 37]; or (iv) **data-driven**, such as clustering or ICA that derive ROIs from the data itself [20, 84]. Subject-specific parcellations further tailor ROIs to individual topography and can enhance phenotype prediction, albeit with added harmonization challenges [55, 93]. Choice of parcellation alone can substantially alter downstream network properties, so reporting and justifying it is essential [4].

3.3 Edge Definition

Edges quantify inter-regional coupling and largely determine graph topology and downstream conclusions. Small changes in connectivity definition can alter reliability and predictive performance, making edge construction a primary driver of result variability [60, 73]. This matters acutely for graph neural networks (GNNs), whose inputs inherit whatever information (or artifacts) the edges encode [57].

We distinguish **static** from **dynamic** connectivity. *Static* methods aggregate the whole scan into a single graph: Pearson correlation (ubiquitous baseline) [11, 86]; partial correlation/precision for conditional associations [65]; nonlinear information measures (e.g., mutual information) [103]; and frequency-domain metrics (e.g., coherence) [88]. *Dynamic* methods estimate time-varying graphs: sliding-window correlations reveal recurring connectivity states [1, 49] but depend on windowing choices; model-based approaches such as dynamic conditional correlation (DCC) avoid arbitrary windows and yield smoothly varying estimates [58]. Reviews synthesize benefits and caveats of dFC and ongoing debates about its added value [62, 75].

In practice, static graphs remain common and effective for many tasks (including GNN-based classification) [57], whereas dynamic graphs can capture state transitions that aid discrimination in some settings. The key is alignment between scientific goal and edge model: linear vs nonlinear, bivariate vs multivariate, single-graph vs time-resolved—each choice reshapes what the graph (and any model trained on it) can learn.

3.4 Filtering Schemes

Connectivity measures often yield dense matrices replete with weak or spurious connections, necessitating filtering to produce a sparse, biologically meaningful network. Despite its importance, no universal standard exists for filtering, and various schemes have emerged, each with distinct implications for network properties [61]. Here, we outline key approaches:

- **Absolute Thresholding**: Retains edges exceeding a fixed weight threshold. While straightforward, it may yield networks with inconsistent densities across subjects.
- **Proportional Thresholding**: Preserves a fixed proportion of the strongest edges, ensuring comparable network densities and facilitating group comparisons [48].
- **Efficiency Cost Optimization**: Balances network efficiency and connectivity cost, reflecting biological constraints [24].
- **Orthogonal Minimum Spanning Trees (OMST)**: Retains critical connections while minimizing redundancy, preserving hierarchical structure [26].
- **Binarization**: Converts weighted edges to binary values, simplifying analysis but sacrificing information on connection strength.

The selected filtering scheme profoundly influences the graph's topological features and the outcomes of subsequent analyses. For instance, proportional thresholding is favored for maintaining consistency across subjects, whereas OMST excels in highlighting core network architecture.

4 Taxonomy

The construction of functional brain graphs from fMRI data involves a series of methodological choices that significantly influence the resulting network's structure and interpretability. This section presents

a detailed taxonomy of the primary components involved in graph construction: node definition, connectivity computation, and filtering schemes. Each category is explored in depth, with a focus on representative methods, their features, and their implications for network analysis. This taxonomy is informed by an extensive review of the literature, ensuring a thorough and up-to-date overview suitable for a peer-reviewed publication.

4.1 Node

Nodes in functional brain graphs typically represent regions of interest (ROIs), which are derived through parcellation of the brain into discrete areas. The choice of parcellation scheme is critical, as it determines the spatial resolution and functional relevance of the nodes, thereby impacting the network's topology and the consistency of connectivity measures across subjects [87]. Here, we categorize and describe three representative approaches to node definition, each with distinct features and applications:

- **Anatomical Parcellation**: This method relies on structural brain regions defined by anatomical atlases. A widely used example is the Automated Anatomical Labeling (AAL) atlas, specifically AAL116, which divides the brain into 116 regions based on anatomical landmarks [91]. AAL116 is favored for its consistency across subjects and ease of integration with structural imaging modalities. However, it may lack functional specificity, as anatomical boundaries do not always align with functional ones.
- **Functional Parcellation**: Functional parcellations are derived from resting-state or task-based fMRI data and aim to capture functional connectivity patterns. The Schaefer100 atlas, for instance, parcellates the brain into 100 regions based on local gradients of functional connectivity [81]. This approach is particularly useful for studies focusing on network dynamics and offers higher functional resolution. However, it may exhibit variability across individuals, complicating group-level analyses.
- **Data-Driven Parcellation**: These methods employ unsupervised techniques, such as independent component analysis (ICA) or clustering, to define nodes based on the data itself. The Power264 atlas, comprising 264 regions, exemplifies this category, derived from clustering resting-state fMRI data [74]. Data-driven approaches adapt to individual variability, potentially enhancing the precision of network models, but they require robust computational resources and may lack standardization across studies.

Each parcellation scheme has unique strengths and limitations. Anatomical parcellations are well-suited for structural studies and group analyses, while functional and data-driven methods are preferable for investigations of dynamic connectivity and personalized modeling, respectively.

4.2 Connectivity as Edge

Connectivity computation defines the edges of the graph, reflecting statistical associations between the time series of node pairs. Our survey identified over 20 distinct methods for computing connectivity, which we categorize into three primary groups for clarity and detailed exploration: pairwise correlation, higher-order functional connectivity (HOFC), and end-to-end models.

4.2.1 Pairwise Correlation

Pairwise correlation is the most prevalent method for defining functional connectivity, particularly the Pearson correlation coefficient, which assesses linear dependencies between the blood-oxygen-level-dependent (BOLD) time series of node pairs. This approach is computationally efficient, straightforward to implement, and widely used due to its interpretability. For example, studies often compute the correlation matrix across ROIs and apply thresholding to form a sparse graph [17].

Despite its popularity, pairwise correlation captures only linear relationships, potentially missing non-linear dynamics that are crucial for understanding complex brain functions. To address this limitation, recent advancements include the use of partial correlation, which controls for indirect connections by isolating direct interactions [85]. Partial correlation is particularly useful for disentangling network hubs but requires larger sample sizes to ensure stability.

4.2.2 Higher-Order Functional Connectivity (HOFC)

HOFC methods extend beyond pairwise correlations to capture nonlinear or higher-order interactions between brain regions. These approaches are increasingly utilized in studies of brain dynamics and complex network states. HOFC methods have been instrumental in modeling complex network states, such as those observed during cognitive tasks or in disease classification [76]. However, interpreting HOFC results can be challenging, often requiring advanced statistical tools and careful consideration of noise sensitivity.

4.2.3 End-to-End Models

End-to-end models represent a recent paradigm shift in connectivity computation, integrating graph construction with downstream prediction tasks through machine learning frameworks. These models do not solely rely on image data but incorporate additional context, such as patient conditions and labels (e.g., disease states), to learn optimal connectivity measures tailored to specific applications. For instance, graph neural networks (GNNs) can embed connectivity matrices into nodes, learning from both the graph structure and external features to enhance classification performance [54].

End-to-end models are particularly effective in personalized medicine, where graphs are adapted to individual profiles. However, they require large datasets for training and may lack interpretability compared to traditional methods. Research indicates that the performance of these models heavily depends on the quality of the input graph, underscoring the importance of careful edge definition [66].

4.3 Filtering Schemes

The initial connectivity matrices derived from fMRI data are often dense, containing numerous weak or spurious connections that may obscure biologically meaningful patterns. Research suggests that dense networks do not necessarily improve analytical performance and that filtering is essential to enhance graph quality by retaining only the most significant connections [61].

The choice of filtering scheme profoundly influences the graph's topological features and the outcomes of subsequent analyses. Proportional thresholding is often preferred for its ability to standardize network density, while OMST is valued for its focus on core structural elements. Despite the variety of methods, no gold standard exists, and the selection should align with the specific research objectives, as underscored by systematic reviews [61].

4.4 Static and Dynamic Graphs

In the construction of functional brain graphs from fMRI data, the distinction between static and dynamic graphs is pivotal, reflecting different approaches to capturing brain connectivity over time. This subsection elaborates on these two categories, their methodologies, and their implications, building on the taxonomy of graph construction methods discussed in Section 4. The analysis is grounded in extensive literature, ensuring a comprehensive overview suitable for a peer-reviewed publication.

4.4.1 Static Functional Brain Graphs: A Time-Invariant Representation

Static functional brain graphs are constructed by utilizing the entire duration of the fMRI time series to compute connectivity measures between nodes. This approach yields a time-invariant representation of functional connectivity, encapsulating the average interaction between brain regions over the scanning period. The process aligns with the general workflow outlined in Section 3 and can be detailed as follows:

1. **Node Definition**: Regions of interest (ROIs) are defined using parcellation schemes such as AAL116, Schaefer100, or Power264, as discussed in Section 4.1. These ROIs serve as nodes, with their time series extracted from the entire fMRI dataset.

2. **Connectivity Computation**: Connectivity measures, such as Pearson's correlation coefficient, are calculated between the time series of each pair of ROIs, using the full duration of the scan. This results in a single connectivity matrix representing the average connectivity across time. Other measures, such as partial correlation or mutual information, may also be employed depending on the research objectives [17].

3. **Filtering Schemes**: The connectivity matrix is then filtered to remove weak or spurious connections, using methods such as absolute thresholding (retaining edges above a fixed value) or proportional thresholding (retaining a fixed percentage of the strongest connections). This step ensures a sparse, biologically meaningful network, as discussed in Section 4.3 [48].

Static graphs are widely used due to their simplicity and ability to capture overall network properties, making them suitable for studies aiming to characterize baseline brain organization, such as small-worldness or modularity [87]. However, they may overlook temporal fluctuations in connectivity, which are critical for understanding dynamic brain states.

4.4.2 Dynamic Functional Brain Graphs: Capturing Temporal Dynamics

to do: summarize more on the dynamic brain graphs. and put them here.

5 Pairwise correlation

In this section, we focus on methods that directly calculate pairwise correlations between regions of interest (ROIs) in the brain, which is a fundamental approach for assessing functional connectivity. These methods operate by evaluating the relationship between the time-series data of two ROIs, producing a correlation value that quantifies their functional interaction. The distinguishing feature of these techniques is their straightforward, direct computation of correlation between two ROIs, without the need for intermediate transformations or complex modeling.

In this paper, we represent the functional connectivity of the brain using an $n \times n$ connectivity matrix, where n denotes the number of regions of interest (ROIs), and each matrix element r_{ij} represents the Pearson correlation coefficient between the time-series data of ROI i and ROI j. The raw data is structured as a $t \times n$ matrix, where t is the number of time points and each column corresponds to an ROI's time-series. The resulting $n \times n$ matrix is symmetric, with diagonal elements $r_{ii} = 1$ representing the perfect self-correlation of each ROI.

5.1 Pearson's Correlation

Pearson's correlation is a widely used method for quantifying the linear relationship between two variables, in this case, the time-series data from two regions of interest (ROIs) in the brain. The Pearson correlation coefficient, denoted by r_{xy}, is defined as:

$$r_{xy} = \frac{\text{cov}(x, y)}{\sqrt{\text{var}(x) \cdot \text{var}(y)}} = \frac{\sum_{i=1}^{t}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{t}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{t}(y_i - \bar{y})^2}} \tag{1}$$

In this equation, $\text{cov}(x, y)$ represents the covariance between the time-series x and y, while $\text{var}(x)$ and $\text{var}(y)$ are the variances of x and y, respectively. This formulation ensures that r_{xy} lies between -1 and 1, where $r_{xy} = 1$ indicates a perfect positive correlation, $r_{xy} = -1$ indicates a perfect negative correlation, and $r_{xy} = 0$ suggests no linear relationship between the time-series.

where covariance and variance are:

$$\text{cov}(x, y) = \frac{1}{t}\sum_{i=1}^{t}(x_i - \bar{x})(y_i - \bar{y}) \tag{2}$$

$$\text{var}(x) = \frac{1}{t}\sum_{i=1}^{t}(x_i - \bar{x})^2, \quad \text{var}(y) = \frac{1}{t}\sum_{i=1}^{t}(y_i - \bar{y})^2 \tag{3}$$

Here, x_i and y_i denote the values of the time-series for the two ROIs at time point i, and $\bar{x}$ and $\bar{y}$ are the mean values of the time-series x and y, respectively:

$$\bar{x} = \frac{1}{t}\sum_{i=1}^{t} x_i, \quad \bar{y} = \frac{1}{t}\sum_{i=1}^{t} y_i \tag{4}$$

Thus, the Pearson correlation coefficient captures the degree of linear association between the two time-series, providing an $n \times n$ connectivity matrix that characterizes the functional connectivity between n ROIs. Despite its simplicity and computational efficiency, Pearson's correlation assumes linearity, which may limit its effectiveness in capturing more complex or non-linear relationships in brain connectivity.

5.2 Cosine Similarity

Cosine similarity is the measure of similarity between two non-zero vector. It determines the degree to which two vectors are pointing in the same direction by calculating the cosine of the angle between them.

To construct a graph representing functional connectivity, we begin with a raw data matrix of shape $t \times n$, where t is the number of time points and n is the number of ROIs. The columns of the matrix represent the time-series data of individual ROIs.

We compute the pairwise cosine correlations between the time series of all ROIs as edges, with a node corresponding to each ROI i, thus constructing a graph.

$$\text{cosine similarity} = S_C(A, B) = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} \tag{5}$$

In this equation, given two ROI time series A and B, $A \cdot B$ represents the dot product of sequence A and sequence B, and $\|A\|\|B\|$ represents the product of the modulus length of sequence A and sequence B.

The value range of cosine similarity is between [-1,1]. When the cosine similarity of two sequences is closer to 1, it means that they are more similar, the two time series have the same direction; on the contrary, the closer it is to -1, it means that they are completely opposite; the closer the similarity is to 0, it means that they have no similarity.

Cosine similarity is more robust to outliers because it is based on the direction of the time series rather than their length. This is especially important when dealing with noisy data. At the same time, cosine similarity performs well when dealing with high-dimensional data. In bioinformatics, it is often used for genome similarity analysis and protein sequence similarity analysis.

5.3 K-Nearest Neighbors (KNN)

To construct a graph representing functional connectivity, we begin with a raw data matrix of shape $t \times n$, where t is the number of time points and n is the number of ROIs. The columns of the matrix represent the time-series data of individual ROIs. Using the K-Nearest Neighbors (KNN) algorithm, we construct a graph where nodes correspond to ROIs, and edges are determined by the pairwise Euclidean distances between their time-series.

For each ROI i, we compute the Euclidean distances between its time-series and those of all other ROIs, identifying the K nearest neighbors with the smallest distances. This relationship is represented by an adjacency matrix $A \in \mathbb{R}^{n \times n}$, where:

$$A_{ij} = \begin{cases} d_{ij} & \text{if } j \text{ is one of the } K \text{ nearest neighbors of } i \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

Here, d_{ij} represents the Euclidean distance between the time-series of ROI i and ROI j. The graph is constructed without self-loops, so $A_{ii} = 0$ for all i. This adjacency matrix captures the local neighborhood structure of the ROIs, with edge weights reflecting the dissimilarity between connected nodes, as measured by Euclidean distance.

5.4 Euclidean Distance

To construct a graph representing functional connectivity, we begin with a raw data matrix of shape $t \times n$, where t is the number of time points and n is the number of ROIs. The columns of the matrix represent the time-series data of individual ROIs. We compute the pairwise Euclidean distances between the time-series of all ROIs to construct a symmetric distance matrix $D \in \mathbb{R}^{n \times n}$, where each entry D_{ij} is defined as:

$$D_{ij} = \|x_i - x_j\| = \sqrt{\sum_{k=1}^{t} (x_{k,i} - x_{k,j})^2} \tag{7}$$

Here, x_i and x_j denote the time-series of ROIs i and j, respectively. The diagonal elements of the matrix are zero, $D_{ii} = 0$, as the distance between an ROI and itself is zero. The matrix is symmetric, satisfying $D_{ij} = D_{ji}$ for all i and j.

This distance matrix provides a comprehensive measure of dissimilarity between all pairs of ROIs, forming the foundation for further graph-based analyses, such as thresholding or clustering, to identify functional connectivity patterns.

5.5 Spearman's Correlation

Spearman's correlation is a non-parametric measure of the monotonic relationship between two variables, often used to capture non-linear associations. In the context of functional connectivity, it is applied to the time-series data from two regions of interest (ROIs). Unlike Pearson's correlation, which measures linear relationships, Spearman's correlation operates on the rank-transformed data, making it robust to outliers and non-normal distributions. The Spearman correlation coefficient, denoted by r_s, is computed as:

$$r_s = 1 - \frac{6\sum_{i=1}^{t} d_i^2}{t(t^2-1)} \tag{8}$$

where d_i is the difference in ranks between the time-series data of ROI i and ROI j for each time point i, and t is the number of time points.

To calculate r_s, the time-series data of each ROI are first rank-transformed. For two time-series x and y, the ranks R_x and R_y are determined, and $d_i = R_{x_i} - R_{y_i}$. The resulting coefficient r_s ranges between -1 and 1, where $r_s = 1$ indicates a perfect monotonic relationship, $r_s = -1$ indicates a perfect inverse monotonic relationship, and $r_s = 0$ suggests no monotonic relationship.

This approach is particularly well-suited for scenarios where the relationships between ROIs may not be strictly linear, offering a more generalizable measure of functional connectivity.

5.6 Kendall's Correlation

Kendall's correlation, also known as Kendall's tau coefficient, is a non-parametric measure of the strength and direction of the monotonic relationship between two variables. It assesses the similarity of the orderings of data when ranked, making it particularly robust to outliers and suitable for time-series data with non-linear relationships. In the context of functional connectivity, Kendall's correlation is applied to the time-series data of two regions of interest (ROIs). The Kendall's tau coefficient, denoted by τ, is defined as:

$$\tau = \frac{C - D}{\frac{1}{2}t(t-1)} \tag{9}$$

where C is the number of concordant pairs, D is the number of discordant pairs, and t is the number of time points. A pair of observations (x_i, y_i) and (x_j, y_j) is concordant if the ranks for both variables agree, meaning $(x_i - x_j)(y_i - y_j) > 0$, and discordant if they disagree, meaning $(x_i - x_j)(y_i - y_j) < 0$.

The coefficient τ ranges from -1 to 1, where: - $\tau = 1$: A perfect agreement in rankings (strong positive monotonic relationship). - $\tau = -1$: A perfect disagreement in rankings (strong negative monotonic relationship). - $\tau = 0$: No consistent agreement or disagreement in rankings (no monotonic relationship).

Kendall's correlation is particularly advantageous for small datasets or when the data distribution is unknown, providing a robust alternative to Pearson's and Spearman's correlation measures in functional connectivity analysis.

5.7 Partial Correlation

Partial correlation quantifies the direct relationship between two variables while accounting for the influence of one or more additional variables. In functional connectivity analysis, partial correlation is particularly useful for identifying direct interactions between regions of interest (ROIs) by removing the effects of other ROIs. For two ROIs, X and Y, and a set of control variables $\mathbf{Z}$, the partial correlation coefficient $r_{XY.\mathbf{Z}}$ is computed as:

$$r_{XY.\mathbf{Z}} = \frac{r_{XY} - r_{XZ}r_{YZ}}{\sqrt{(1-r_{XZ}^2)(1-r_{YZ}^2)}} \tag{10}$$

where r_{XY}, r_{XZ}, and r_{YZ} are the Pearson correlation coefficients between the corresponding variables. This formula provides an exact partial correlation when controlling for a single variable Z. For multiple

control variables, partial correlation is derived from the inverse covariance matrix $\mathbf{\Sigma}^{-1}$, also known as the precision matrix, as:

$$r_{ij.\mathbf{Z}} = -\frac{\mathbf{\Sigma}_{ij}^{-1}}{\sqrt{\mathbf{\Sigma}_{ii}^{-1} \cdot \mathbf{\Sigma}_{jj}^{-1}}} \tag{11}$$

Here, $\mathbf{\Sigma}_{ij}^{-1}$ represents the (i, j)-th element of the inverse covariance matrix.

The intuition behind partial correlation is to isolate the direct association between X and Y by first removing the effects of $\mathbf{Z}$ through linear regression. The residuals from these regressions, r_X and r_Y, capture the variance of X and Y not explained by $\mathbf{Z}$, and the correlation of these residuals reflects their partial relationship.

In functional connectivity, partial correlation is often used to construct sparse brain networks, highlighting direct functional interactions while accounting for the influence of other regions. This method is advantageous in reducing spurious connections caused by global or indirect effects.

5.8 Wavelet Coherence

Wavelet coherence is a powerful frequency domain analysis technique used to measure the coherence between two time series signals. It does this by decomposing the time series into wavelet coefficients and then calculating the coherence between the wavelet coefficients. Wavelet coherence can reveal the coherence relationship between time series as a function of time and frequency.

In complex time series analysis, especially when signals exhibit significant non-stationarity, traditional linear correlation analysis methods cannot reveal the true coherence between time series signals. Wavelet coherence is able to find temporally locally coherent oscillatory behavior and interdependencies in two time series at different time scales.

The wavelet coherence of two time series x and y is:

$$W_{xy}(s) = \frac{|S(W_x^*(s)W_y^*(s))|}{S(|W_x(s)|^2)S(|W_y(s)|^2)} \tag{12}$$

where $W_x(s)$ and $W_y(s)$ denote the continuous wavelet transforms of x and y at frequency s. The superscript $*$ is the complex conjugate, and S is a smoothing operator in time and scale. For real-valued time series, the wavelet coherence is real valued if you use a real-valued analyzing wavelet, and complex-valued if you use a complex-valued analyzing wavelet.

Wavelet coherence is widely used in the field of signal processing. It can be used to process non-stationary signals, such as biomedical signals, seismic signals, and can extract frequency information and time-varying features. Wavelet coherence is often used in electroencephalograms. For the frequency distribution of unknown signals, good resolution can be obtained on the time axis, which is suitable for unstable signal analysis and processing of brain waves.

5.9 Cityblock Distance

Cityblock distance is also called "Manhattan distance". As the name implies, if you drive from one intersection to another in Manhattan, the driving distance is obviously not the straight-line distance between the two points, but the distance between the two points in the north-south direction plus the distance in the east-west direction. This actual driving distance is the cityblock distance.

Cityblock distance is derived by looking at the difference in the absolute value of each dimension in two time series signals, representing the difference between them. It is given by the following formula:

$$d(\mathbf{A}, \mathbf{B}) = \sum_{i=1}^{n} |A_i - B_i| \tag{13}$$

where A_i and B_i represent two time series signals. From the formula definition, the cityblock distance is always greater than or equal to zero. For the same time series signal, the similarity is zero, which is the same as the Euclidean distance.

In most cases, the meaning of the cityblock distance and the Euclidean distance are similar, both of which are used to describe the similarity between two time series. However, the city block distance only needs addition and subtraction, which makes it less expensive in a large number of calculations, suppresses the influence of large differences in a single dimension in the Euclidean distance calculation

process (no square operation is required), and eliminates the error caused by taking approximate values in the square root process.

5.10 Dynamic Time Warping

Dynamic time warping(DTW) is an algorithm that uses dynamic programming strategies to dynamically warp incompletely synchronized sequences (usually time series) to correctly calculate the maximum possible similarity between them.

This is achieved by warping the time axis of the sequence to minimize the distance between corresponding points on the sequence. Unlike traditional distance metrics such as Euclidean distance, it can align sequences of different lengths. It is often used to measure the distance between two time series.

Formally, DTW distance can be defined as:

$$DTW(A, B) = \min \sqrt{\sum_{i,j} d(a_i, b_j)^2} \tag{14}$$

where a_i and b_i represent two time series signals.

At its core, DTW is an algorithm for aligning and comparing two time series datasets. Unlike simple methods that compare points based on their position in the time series, DTW focuses on the shape of the data. This enables it to compare and align time series that may be out of sync or of different lengths through non-linear alignment.

DTW is computationally expensive, especially for long sequences, as the complexity of the algorithm is quadratic in the length of the time series. And if DTW is unconstrained, it may over-distort the sequence, resulting in unnatural alignments.

5.11 Earth Mover's Distance

Earth Mover's Distance (EMD), also known as Wasserstein distance, is a dissimilarity measure that assumes that each signal is a probability distribution and represents the minimum cost of converting one time series signal into another time series. It can also be used to represent the similarity between two time series.

We regard the two time series x and y as two piles of soil. If one pile of soil is moved to the position and shape of another pile of soil, there are many possible solutions. The cost of moving soil is defined as the amount of soil moved multiplied by the distance the soil is moved. Among all the solutions, the minimum average distance required to move from sequence x to sequence y is the solution with the minimum cost of moving soil. This cost is called the Wasserstein distance between the two sequences.

$$x = \{(t_{x_1}, x_1), (t_{x_2}, x_2), \ldots, (t_{x_m}, x_m)\} \tag{15}$$

$$y = \{(t_{y_1}, y_1), (t_{y_2}, y_2), \ldots, (t_{y_n}, y_n)\} \tag{16}$$

where each x_i is a cluster (=amplitude) of the signal x at time-point t_{x_i} and each y_j is a cluster of signal y at time-point t_{y_j} .

Earth mover's distance can be calculated for signals of different lengths. In the case of BOLD signals, for a given individual, although the sequence lengths m and n are not necessarily equal in general, they can be considered to be the same and the lengths scanned directly.Then the ground distance between clusters at p_j and q_j can be encoded in the matrix

$$D = \left[d_{i,j}\right] \tag{17}$$

with a flow between clusters at pi and qj represented by the matrix,

$$F = \left[f_{i,j}\right] \tag{18}$$

The objective is to minimize the overall cost,

$$\min \sum_{i=1}^{m} \sum_{j=1}^{n} f_{i,j} d_{i,j} \tag{19}$$

while satisfying the following constraints:

$$f_{i,j} \geq 0 \text{ for } 1 \leq i \leq m,\ 1 \leq j \leq n \tag{20}$$

$$\sum_{j=1}^{n} f_{i,j} \leq t_{x_i} \text{ for } 1 \leq i \leq m \tag{21}$$

$$\sum_{i=1}^{m} f_{i,j} \leq t_{y_j} \text{ for } 1 \leq j \leq n \tag{22}$$

$$\sum_{i=1}^{m}\sum_{j=1}^{n} f_{i,j} = \min\left\{\sum_{i=1}^{m} t_{x_i}, \sum_{j=1}^{n} t_{y_j}\right\} \tag{23}$$

Earth mover's distance can then be defined as the amount of work needed to transform time series x to time series y, normalized by the total flow.

$$EMD(x,y) = \frac{\sum_{i=1}^{m}\sum_{j=1}^{n} f_{i,j} d_{i,j}}{\sum_{i=1}^{m}\sum_{j=1}^{n} f_{i,j}} \tag{24}$$

Similar to DTW, EMD considers the nonlinear interactions between time series signals, has scale variability, and is applicable to sequences of generally unequal lengths.

5.12 Mutual Information

Mutual Information (MI) is an information-theoretic metric that quantifies the amount of shared information between two variables, such as the time-series data from two regions of interest (ROIs) in the brain. Unlike Pearson's correlation, MI captures both linear and non-linear dependencies, making it a versatile tool for assessing complex relationships in functional connectivity.

The mutual information $I(X;Y)$ between two time-series X and Y is defined as:

$$I(X;Y) = \int\int p(x,y) \log\left(\frac{p(x,y)}{p(x)p(y)}\right) dx\, dy \tag{25}$$

where $p(x,y)$ represents the joint probability density function of X and Y, and $p(x)$ and $p(y)$ are the marginal probability density functions of X and Y, respectively. In practice, these probability densities are often estimated using techniques such as kernel density estimation or histograms.

Mutual information can also be expressed in terms of entropy, providing a clearer interpretation:

$$I(X;Y) = H(X) + H(Y) - H(X,Y) \tag{26}$$

where:

- $H(X) = -\int p(x) \log p(x)\, dx$: Entropy of X, measuring the uncertainty of X.
- $H(Y) = -\int p(y) \log p(y)\, dy$: Entropy of Y, measuring the uncertainty of Y.
- $H(X,Y) = -\int\int p(x,y) \log p(x,y)\, dx\, dy$: Joint entropy of X and Y, measuring their combined uncertainty.

The mutual information captures the amount of information X and Y share, with $I(X;Y) = 0$ if and only if X and Y are independent. This property makes it particularly valuable for identifying both linear and non-linear relationships.

In this paper, the functional connectivity between ROIs is represented by an $n \times n$ connectivity matrix, where each element m_{ij} represents the mutual information between the time-series data of ROI i and ROI j. Mutual information is symmetric ($m_{ij} = m_{ji}$), ensuring that the connectivity matrix remains symmetric as well.

Although more computationally demanding than correlation-based methods due to the need for estimating probability densities and performing numerical integration, mutual information provides a powerful and general measure of dependency. This makes it a valuable tool for uncovering complex interactions in functional brain networks that might not be detectable with linear measures alone.

5.13 Coherence

Coherence is a frequency-domain measure that quantifies the degree of phase consistency between two time-series signals, such as those recorded from different regions of interest (ROIs) in the brain. Unlike correlation, which captures linear relationships, coherence is specifically designed to assess the consistency of interactions across specific frequency bands, making it particularly useful for analyzing oscillatory dynamics in functional connectivity. The coherence between two time-series X and Y at a given frequency λ is defined as:

$$Coh_{XY}(\lambda) = \frac{|P_{XY}(\lambda)|^2}{P_{XX}(\lambda)P_{YY}(\lambda)}$$

where $P_{XY}(\lambda)$ is the cross-spectral density between X and Y, which captures the shared frequency components of the two signals, and $P_{XX}(\lambda)$ and $P_{YY}(\lambda)$ are the power spectral densities of X and Y, respectively, describing the distribution of power across frequencies for each signal. Coherence values range between 0 and 1, where $Coh_{XY}(\lambda) = 1$ indicates perfect phase locking at frequency λ, and $Coh_{XY}(\lambda) = 0$ indicates no consistent phase relationship. The symmetry property, $Coh_{XY}(\lambda) = Coh_{YX}(\lambda)$, ensures consistent results regardless of signal order.

In functional connectivity studies, coherence is computed for all pairs of ROIs, generating an $n \times n$ connectivity matrix where each element c_{ij} represents the coherence between the time-series of ROI i and ROI j across a specified frequency range. The input data is typically a $t \times n$ matrix, where t is the number of time points, and each column corresponds to an ROI's time-series. Coherence is often averaged over specific frequency bands of interest (e.g., low-frequency resting-state bands) or computed separately for each frequency to provide a detailed spectral representation of connectivity. Extensions like partial coherence account for the influence of other ROIs in the network, offering refined estimates of connectivity.

Coherence is valuable for exploring frequency-specific interactions and is particularly suited for studying oscillatory dynamics in brain networks. However, it requires careful preprocessing, including band-pass filtering and appropriate spectral estimation, to ensure accuracy. Despite these challenges, coherence complements time-domain measures like Pearson's correlation, providing unique insights into the dynamic and spectral characteristics of functional brain connectivity.

5.14 Granger Causality

Granger causality is a statistical method used to infer directional interactions between time-series data, such as those derived from regions of interest (ROIs) in the brain. It assesses whether the past values of one time-series can enhance the prediction of another time-series beyond what the latter's own history can achieve. This method is particularly effective for analyzing causal relationships in fMRI data, where time-series signals represent the activity of multiple ROIs.

For two time-series X and Y, Granger causality evaluates two models: an unrestricted model, which includes the past values of both X and Y, and a restricted model, which uses only the past values of Y. These models are defined as:

$$Y_t = \sum_{i=1}^{p} a_i Y_{t-i} + \sum_{i=1}^{p} b_i X_{t-i} + \epsilon_t$$

$$Y_t = \sum_{i=1}^{p} a_i Y_{t-i} + \eta_t$$

where Y_t represents the value of Y at time t, a_i and b_i are the model coefficients, p is the model order (number of lags), and ϵ_t and η_t are the residuals (prediction errors) of the unrestricted and restricted models, respectively. The strength of Granger causality is determined by computing the F-statistic:

$$F = \frac{(RSS_r - RSS_u)/p}{RSS_u/(n - 2p - 1)}$$

where RSS_r and RSS_u are the residual sums of squares for the restricted and unrestricted models, respectively, and n is the number of observations. If the F-statistic exceeds a critical value, X is considered to Granger-cause Y. A symmetric analysis can be performed to test whether Y Granger-causes X, enabling the identification of bidirectional or unidirectional causal relationships.

In fMRI functional connectivity, Granger causality is calculated between the time-series of all ROI pairs, resulting in an $n \times n$ directed connectivity matrix, where n is the number of ROIs. Each matrix element g_{ij} represents the Granger causal influence of ROI i on ROI j. Extensions such as conditional Granger causality refine the analysis by controlling for additional variables, while frequency-domain methods, including the Directed Transfer Function (DTF), allow the study of causal interactions within specific frequency bands.

Granger causality provides a robust framework for investigating directed interactions in brain networks. Its ability to capture directional relationships makes it a valuable tool in connectivity studies. However, its assumptions of stationarity and linearity, as well as its sensitivity to noise, necessitate careful preprocessing and validation, particularly in the complex context of fMRI data.

5.15 Generalised Synchronisation

Generalised synchronisation is a nonlinear method used to evaluate the synchrony between two time-series by analyzing their interdependence within reconstructed state spaces. Unlike linear measures such as Pearson's correlation, generalised synchronisation captures complex, nonlinear relationships, making it particularly valuable for functional connectivity analysis in fMRI data, where nonlinear dynamics are prevalent.

The method involves reconstructing the state spaces of two time-series X and Y by embedding them into higher-dimensional spaces using time delays. The reconstructed state spaces for X and Y are defined as:

$$\mathbf{X}(t) = \begin{bmatrix} x(t) & x(t-\tau) & \cdots & x(t-(m-1)\tau) \end{bmatrix}^\top, \quad \mathbf{Y}(t) = \begin{bmatrix} y(t) & y(t-\tau) & \cdots & y(t-(m-1)\tau) \end{bmatrix}^\top$$

where m is the embedding dimension, τ is the time delay, and $\mathbf{X}(t)$ and $\mathbf{Y}(t)$ are the embedded vectors representing the reconstructed states of X and Y, respectively. These reconstructed state spaces allow the evaluation of interdependence between the time-series by comparing their local dynamics in the state space.

To quantify synchronisation, directional measures are computed, assessing whether changes in one time-series predict changes in the other. These measures are based on the similarity of neighborhoods in the reconstructed spaces, providing directional and asymmetric information about the interaction. For instance, the directional influence from X to Y can be quantified separately from Y to X, highlighting potential driver-response relationships between the signals.

For fMRI functional connectivity analysis, generalised synchronisation is applied to the time-series of all regions of interest (ROIs), resulting in an $n \times n$ connectivity matrix, where n is the number of ROIs. Each element of the matrix reflects the degree of synchronisation or directional influence between a pair of ROIs, offering a detailed map of nonlinear interactions across the brain. The parameters m (embedding dimension) and τ (time delay) are critical to the analysis and must be chosen carefully to ensure accurate state space reconstruction. Typical parameter choices involve $m = 10$ and $\tau = 2$, with neighborhood dynamics estimated using 10 nearest neighbors.

Generalised synchronisation provides a powerful framework for exploring complex, nonlinear dynamics in brain networks, complementing traditional linear measures. However, it requires careful parameter tuning and robust preprocessing of the time-series data, including detrending and normalization, to ensure reliable and interpretable results. Its ability to capture both symmetric and directional interactions makes it an invaluable tool for understanding the intricate connectivity patterns underlying brain function.

5.16 Patel's Conditional Dependence

Patel's conditional dependence is a statistical method designed to evaluate the dependency between two time-series by analyzing the imbalance in their conditional probabilities, $P(x|y)$ and $P(y|x)$. This approach is particularly effective for capturing nonlinear relationships and directionality, making it a valuable tool for functional connectivity analysis in fMRI data. The method provides two metrics: κ, which quantifies the strength of the dependency, and τ, which captures the directionality.

The method begins by normalizing the time-series data into the range $[0, 1]$. Data values below the 10th percentile are mapped to 0, values above the 90th percentile are mapped to 1, and values between the 10th and 90th percentiles are linearly scaled. These normalized time-series, X and Y, are then used to calculate conditional probabilities. For each pair of time-series, the probabilities are defined as:

$$P(x|y) = \frac{\text{Count}(X = x, Y = y)}{\text{Count}(Y = y)}, \quad P(y|x) = \frac{\text{Count}(X = x, Y = y)}{\text{Count}(X = x)}.$$

The connection strength, κ, is computed as:

$$\kappa = \frac{1}{2}\left(\sum_{x,y} |P(x|y) - P(x)| + \sum_{x,y} |P(y|x) - P(y)|\right),$$

where $P(x)$ and $P(y)$ are the marginal probabilities of X and Y, respectively. This metric measures the overall dependency between X and Y, regardless of directionality.

The connection directionality, τ, is defined as:

$$\tau = \frac{1}{2}\left(\sum_{x,y}(P(x|y) - P(x)) - \sum_{x,y}(P(y|x) - P(y))\right).$$

This asymmetry quantifies the imbalance in the conditional probabilities and indicates the dominant direction of interaction between X and Y.

In functional connectivity analysis for fMRI data, Patel's conditional dependence is applied to all pairs of regions of interest (ROIs). The resulting connectivity matrix is $n \times n$, where n is the number of ROIs. Each matrix element contains the computed values of κ and τ for the corresponding ROI pair. High κ values indicate strong connectivity, while nonzero τ values provide insights into the direction of interaction.

Patel's method allows optional binarization of the normalized data at thresholds such as 0.25, 0.5, or 0.75. While no binarization or thresholds near 0.75 often yield the most robust results, the choice of threshold affects the sensitivity and specificity of the analysis. Proper preprocessing, including normalization and optional binarization, is essential for reliable results.

By incorporating both connection strength and directionality, Patel's conditional dependence offers a nuanced framework for understanding functional connectivity. Its ability to capture nonlinear dependencies and directional interactions makes it a powerful tool for studying complex brain networks.

5.17 LiNGAM

LiNGAM (Linear Non-Gaussian Acyclic Model) is a method for discovering causal relationships between regions of interest (ROIs) in the brain, based on observed time-series data. Unlike pairwise correlation methods, which focus only on statistical relationships, LiNGAM identifies directional influences between variables under the assumption of linearity and non-Gaussianity of the disturbance terms.

To construct the causal structure of functional connectivity, LiNGAM assumes a directed acyclic graph (DAG) where each observed variable (ROI time series) is modeled as a linear combination of its parent nodes (causal ROIs) and an independent noise term. Mathematically, the model is expressed as:

$$x_i = \sum_{j=1}^{n} b_{ij} x_j + e_i, \quad i = 1, 2, ..., n, \quad j < i, \tag{27}$$

where:

- x_i represents the time series of ROI i,
- b_{ij} are the causal coefficients quantifying the influence of ROI j on ROI i, where $j < i$ represents the causal order,
- e_i is the disturbance (noise) term, which is independent, non-Gaussian, and has non-zero variance.

The fundamental assumptions of LiNGAM are:

1. The causal structure can be represented as a directed acyclic graph (DAG).
2. The relationships between ROIs are linear.
3. The disturbance terms e_i are mutually independent and follow non-Gaussian distributions.

Given the observed time-series data, structured as a $t \times n$ matrix (where t is the number of time points and n is the number of ROIs), LiNGAM operates as follows:

1. First, subtract the mean from each time series to center the data:

$$\bar{x}_i = \frac{1}{t}\sum_{k=1}^{t} x_{ik}, \quad \text{and replace } x_{ik} \text{ with } x_{ik} - \bar{x}_i. \tag{28}$$

2. Apply Independent Component Analysis (ICA) to estimate the linear mixing matrix A, which satisfies:

$$\mathbf{x} = A\mathbf{e}, \quad A = (I - B)^{-1}, \tag{29}$$

   where $\mathbf{x}$ is the matrix of observed time-series data, $\mathbf{e}$ is the vector of independent noise terms, and B is the causal coefficient matrix.

3. Resolve the permutation and scaling ambiguities inherent in ICA to obtain the correct causal ordering of the variables.

4. Derive the causal coefficient matrix B by normalizing the estimated A.

The resulting $n \times n$ matrix B contains directed edges representing causal influences between ROIs, where a non-zero element b_{ij} indicates a directed causal relationship from ROI j to ROI i. Unlike symmetric correlation matrices, the LiNGAM-derived B is asymmetric due to its directional nature.

5.18 PLV

Phase Locking Value (PLV) is a widely used metric for analyzing phase synchronization between two signals. It is particularly effective in assessing the functional connectivity of brain regions by quantifying the consistency of phase differences between time-series data. Unlike methods that rely solely on amplitude or linear relationships, PLV focuses exclusively on phase relationships, making it robust to amplitude variations.

PLV is defined as the magnitude of the average phase difference between two signals, $x(t)$ and $y(t)$, over a series of time points or trials. Mathematically, it can be expressed as:

$$plv_{xy}(\omega) = \frac{\left|\frac{1}{n}\sum_{k=1}^{n} 1_x(\omega,k)1_y(\omega,k)e^{i(\varphi_x(\omega,k)-\varphi_y(\omega,k))}\right|}{\sqrt{\left(\frac{1}{n}\sum_{k=1}^{n} 1_x^2(\omega,k)\right)\left(\frac{1}{n}\sum_{k=1}^{n} 1_y^2(\omega,k)\right)}} \tag{30}$$

Here, $1_x(\omega,k)$ and $1_y(\omega,k)$ represent the amplitudes of the time series signals x and y at frequency ω. $\varphi_x(\omega,k)$ and $\varphi_y(\omega,k)$ denote the phases of the signals x and y at frequency ω and time point k, respectively. $e^{i(\varphi_x(\omega,k)-\varphi_y(\omega,k))}$ represents the complex form of the phase difference between the two signals, which is used to measure the degree of phase synchronization.

$$plv_{xy}(\omega) = \left|\frac{1}{n}\sum_{k=1}^{n} e^{i(\varphi_x(\omega,k)-\varphi_y(\omega,k))}\right| \tag{31}$$

The formula is a simplified version, which ignores the effect of amplitude and focuses solely on the synchronization of phase differences, directly reflecting the degree of phase synchronization.

PLV provides a measure of phase coherence between signals, which is crucial for understanding neural interactions. For example, in EEG studies, PLV has been applied to investigate communication between brain regions during cognitive tasks, motor control, and sensory processing. By focusing solely on phase information, PLV avoids potential confounds introduced by amplitude variability, making it a suitable tool for analyzing signals with varying power levels.

5.19 PSI

The phase slope index (PSI) is a more general quantity for inferring the one-way interaction of two time series data (Nolte et al., 2008). It is calculated from the complex-valued coherence and quantifies the consistency of the direction of change of the phase difference over frequency between different series.

In simple terms, if the amplitudes of changes in different time series are similar, then the phase difference between the two series will increase with increasing frequency, resulting in a positive slope of the phase difference.

The proposed method is based on the slope of the cross-spectral phase between two time series. The metric is the "Phase Slope Index" (PSI) and is defined as

$$\tilde{\Psi}_{ij} = \Im\left(\sum_{f\in F} C^*_{ij}(f)C_{ij}(f+\delta f)\right) \tag{32}$$

where,

$$C_{ij}(f) = \frac{S_{ij}(f)}{\sqrt{S_{ii}(f)S_{jj}(f)}} \tag{33}$$

is the complex coherency, S is the cross-spectral matrix between two time series data, δf is the frequency resolution, and $\Im(\cdot)$ denotes taking the imaginary part, F is the set of frequencies over which the slope is summed.To see that the definition of $\tilde{\Psi}_{ij}$ corresponds to a meaningful estimate of the average slope it is convenient to rewrite it as

$$\tilde{\Psi}_{ij} = \sum_{f\in F} \alpha_{ij}(f)\alpha_{ij}(f+\delta f)\sin(\Phi(f+\delta f)-\Phi(f)) \tag{34}$$

with $\alpha_{ij}(f) = |C_{ij}(f)|$ being frequency dependent weights. For smooth phase spectra, $\sin(\Phi(f+\delta f)-\Phi(f)) \approx \Phi(f+\delta f)-\Phi(f)$ and hence Ψ corresponds to a weighted average of the slope. We emphasize that since Ψ vanishes if the imaginary part of coherency vanishes it will will therefore not be robust to mixing of non-interacting sources.

Finally, it is convenient to normalize $\bar{\Psi}$ by an estimate of its standard deviation

$$\Psi = \frac{\tilde{\Psi}}{\mathrm{std}(\tilde{\Psi})} \tag{35}$$

Positive values of the phase slope index indicate that the first sequence signal leads the second sequence signal. The phase slope index is a bivariate measure that quantifies the consistency of the phase lag between the two signals with frequency. For a wider frequency range, a value significantly deviates from zero, indicating that one of the signals consistently leads the other, which itself indicates a "causal" (at least time-delayed) interaction between the two signals.

5.20 Distance correlation

Distance correlation can effectively reflect both linear and nonlinear dependencies between two random vectors of any dimension. Specifically, two vectors are independent if and only if their distance correlation is zero [89]. In contrast, Pearson's correlation being zero does not imply independence, as the variables may still have a nonlinear relationship. This makes distance correlation better suited for capturing more complex dependencies than Pearson's correlation.

Assume that $\{a_i\}_{i=1}^n$ and $\{b_i\}_{i=1}^n$ are n paired sample from two random vectors $a \in \mathbb{R}^p$ and $b \in \mathbb{R}^q$, where the dimension p and q are not required to be same. The distance correlation can be defined as four steps [96]:

1. Calculate the Euclidean distance matrices $A \in \mathbb{R}^{n\times n}$ and $B \in \mathbb{R}^{n\times n}$, where $A_{ij} = \| a_i - a_j \|_2$ and $B_{ij} = \| b_i - b_j \|_2$.
2. Calculate the U-centered distance matrices

$$\widehat{A}_{ij} = \begin{cases} A_{ij} - \frac{\Sigma_{l=1}^n A_{il}}{n-2} - \frac{\Sigma_{k=1}^n A_{kj}}{n-2} + \frac{\Sigma_{k,l=1}^n A_{kl}}{(n-1)(n-2)}, & i \neq j \\ 0, \quad i = j \end{cases} \tag{36}$$

   For $1 \leq i,\, j \leq n$ and $\widehat{B} \in \mathbb{R}^{n\times n}$

3. Define the distance covariance (dCov) by

$$dCov(a,b) = \frac{\Sigma_{i\neq j}\widehat{A}_{ij}\widehat{B}_{ij}}{n(n-3)} \tag{37}$$

4. Define the distance correlation (dCor) by

$$dCor(a.b) = \sqrt{\frac{dCov(a,b)}{\sqrt{dCov(a,a)dCov(b,b)}}} \tag{38}$$

   if $dCov(a,b) > 0$, and otherwise 0.

6 High-Order Functional Connectivity

HOFC methods extend beyond pairwise correlations of BOLD time series to quantify higher-level relationships among regions of interest (ROIs). Rather than correlating raw signals alone, HOFC correlates *connectivity profiles*—i.e., each region's connectivity to all other regions—to capture organizational principles that are not directly visible at the signal level [100]. These approaches have shown complementary sensitivity to disease-related alterations and provide a useful counterpart to low-order FC (LOFC) [100, 101].

6.1 Correlation's Correlation

Let $X_i \in \mathbb{R}^T$ denote the BOLD time series of ROI i ($i = 1, \ldots, N$) with T time points. The conventional (low-order) functional connectivity matrix $W \in \mathbb{R}^{N\times N}$ is computed via Pearson correlations,

$$W_{ij} = \mathrm{corr}(X_i, X_j), \quad 1 \le i, j \le N.$$

Following [100], we first apply Fisher's r-to-z transform to W to improve normality, yielding Z with entries $Z_{ij} = \frac{1}{2}\ln\left(\frac{1+W_{ij}}{1-W_{ij}}\right)$. For each region i, define its *topographical FC profile* as the column vector $L_i \in \mathbb{R}^{N-2}$ formed by $Z_{\cdot i}$ after removing the two self-terms corresponding to ROIs i and j (to avoid nuisance effects). The *high-order* connectivity between regions i and j is then the Pearson correlation between their FC profiles:

$$\mathrm{HOFC}_{ij} \;=\; \mathrm{corr}\big(L_i, L_j\big) \;=\; \frac{\sum_{k\in\{1,\ldots,N\}\setminus\{i,j\}}(Z_{ki} - \overline{Z_{\cdot i}})(Z_{kj} - \overline{Z_{\cdot j}})}{\sqrt{\sum_{k\neq i,j}(Z_{ki} - \overline{Z_{\cdot i}})^2}\,\sqrt{\sum_{k\neq i,j}(Z_{kj} - \overline{Z_{\cdot j}})^2}}.$$

This "correlation's correlation" yields an HOFC matrix of the same size as W, but it is *not* redundant with LOFC: two regions can have weak LOFC yet strong HOFC if their global connectivity *patterns* are similar, and vice versa [100]. Empirically, HOFC has been shown to increase sensitivity to group differences (e.g., in mild cognitive impairment) relative to LOFC, and its graph-theoretic properties provide complementary information [100].

6.2 dHOFC: Dynamics-Based High-Order FC

Evidence suggests that functional coupling fluctuates over short timescales and relates to ongoing cognition [40, 75]. Dynamics-based HOFC (dHOFC) first computes sliding-window LOFC time series for each edge and then correlates *edge trajectories* to characterize synchrony among connections [106]. Let $w_{ij}(\theta)$ be the LOFC between ROIs i, j within window $\theta = 1, \ldots, \Theta$ (window length L, step s). The dHOFC between two edges (i, j) and (p, q) is

$$\mathrm{dHOFC}_{ij,pq} \;=\; \frac{\sum_{\theta=1}^{\Theta}\big(w_{ij}(\theta) - \overline{w_{ij}}\big)\big(w_{pq}(\theta) - \overline{w_{pq}}\big)}{\sqrt{\sum_{\theta=1}^{\Theta}\big(w_{ij}(\theta) - \overline{w_{ij}}\big)^2}\,\sqrt{\sum_{\theta=1}^{\Theta}\big(w_{pq}(\theta) - \overline{w_{pq}}\big)^2}}.$$

Because dHOFC lives in an $N^2 \times N^2$ edge–edge space, dimensionality reduction (e.g., clustering of edge time series, then averaging within clusters) is commonly used before downstream modeling [19, 106]. dHOFC captures time-resolved high-order organization and has proven useful in disease classification, albeit with somewhat lower reliability than static HOFC unless paradigms enhance stability [101].

6.3 tHOFC: Topographical Profile Similarity

Topographical HOFC (tHOFC) operationalizes the correlation's-correlation idea using the LOFC matrix W (or Z) as input and measures the similarity between two regions' connectivity profiles across the rest of the brain [100, 101, 109]. Let w_{ik} denote the LOFC between ROIs i and k, and let $\overline{w_{i\cdot}}$ be its mean across $k \neq i$. Then:

$$\mathrm{tHOFC}_{ij} \;=\; \frac{\sum_{k\neq i,j}(w_{ik} - \overline{w_{i\cdot}})(w_{jk} - \overline{w_{j\cdot}})}{\sqrt{\sum_{k\neq i,j}(w_{ik} - \overline{w_{i\cdot}})^2}\,\sqrt{\sum_{k\neq i,j}(w_{jk} - \overline{w_{j\cdot}})^2}}.$$

Practical implementations often Fisher-z transform W before profile correlation, omit diagonals, and can include optional nuisance handling. tHOFC reliably captures mesoscale organizational similarity and shows fair-to-good test–retest reliability [101].

6.4 aHOFC: Associated High-Order FC

Associated HOFC (aHOFC) quantifies the *inter-level association* between a region's HOFC topography and another region's LOFC topography, complementing both LOFC and tHOFC [101, 104]. For ROIs i and j, define

$$\text{aHOFC}_{ij} = \frac{\sum_{k\neq i,j}(\text{tHOFC}_{ik} - \overline{\text{tHOFC}_{i\cdot}})\,(w_{jk} - \overline{w_{j\cdot}})}{\sqrt{\sum_{k\neq i,j}(\text{tHOFC}_{ik} - \overline{\text{tHOFC}_{i\cdot}})^2}\,\sqrt{\sum_{k\neq i,j}(w_{jk} - \overline{w_{j\cdot}})^2}}.$$

By explicitly linking the high-order and low-order organizations, aHOFC has shown improved classification performance over LOFC and even tHOFC in certain clinical applications [104], while maintaining fair-to-good reliability [101].

6.5 Sparse HOFC

To capture complex, multivariate interactions that go beyond pairwise correlations, **(author?)** [102] proposed a framework that integrates Bayesian network modeling with sparse representation. This approach assumes that functional brain activities are realized via higher-order interactions among specific networks, which can be inferred as directed acyclic graphs (DAGs) rather than simple undirected correlation graphs.

Let the fMRI data be defined over a set of ROIs V. The time series for each subject are first segmented into blocks corresponding to task paradigms (e.g., task vs. baseline). For each temporal segment d, a Bayesian network structure $G = (V, E)$ is inferred using the Greedy Equivalence Search (GES) algorithm. GES searches for the optimal DAG that maximizes a network score $S(G)$, typically defined by the posterior probability of the graph given the data:

$$S(G) = p(G, d) = p(d|G)p(G),$$

where $p(d|G)$ represents the likelihood of observing the dataset d under the graphical model G (capturing conditional independencies), and $p(G)$ is the prior of the graph. This step generates a set of adjacency matrices representing higher-order interaction patterns for each time segment.

To extract consistent canonical patterns from these varied segments, the method employs an online dictionary learning algorithm for sparse coding. Each inferred interaction graph (adjacency matrix) with m_v ROIs is flattened into a vector $x_i \in \mathbb{R}^m$, where $m = m_v \times m_v$. Given a set of n training samples (graphs) $X = [x_1, \ldots, x_n]$, the goal is to learn a dictionary $D \in \mathbb{R}^{m\times k}$ containing k atoms (basis networks) and a sparse weight vector α_i for each sample, such that $x_i \approx D\alpha_i$. The objective function minimizes the reconstruction error with an ℓ_1 regularization term to induce sparsity:

$$\min_{D,\alpha} \frac{1}{n}\sum_{i=1}^{n}\left(\frac{1}{2}\|x_i - D\alpha_i\|_2^2 + \lambda\|\alpha_i\|_1\right),$$

where λ controls the sparsity level. The resulting dictionary atoms $d_j \in D$ represent the fundamental, recurrent ROI-by-ROI high-order functional interaction patterns characteristic of the brain states during the task [102].

6.6 MVND-Based High-Order FC

To address the limitation of Pearson correlation (PC) in capturing only pairwise statistics, **(author?)** [108] proposed a framework based on the Matrix Variate Normal Distribution (MVND) to estimate both low- and high-order functional connectivity (FC) simultaneously.

Let $X \in \mathbb{R}^{V\times P}$ denote the rs-fMRI data matrix with V time points and P ROIs. First, a sliding window approach is employed to generate a sequence of K temporal subseries. For each window k, a temporal FC network $W^{(k)} \in \mathbb{R}^{P\times P}$ is constructed via Pearson correlation [108]. The collection of these temporal networks is treated as samples from a random matrix variable $\mathcal{W}$ following an MVND:

$$\mathcal{W} \sim N(M, \Omega \otimes \Omega),$$

where $M \in \mathbb{R}^{P\times P}$ represents the population mean (corresponding to the static, low-order FC), and $\Omega \in \mathbb{R}^{P\times P}$ is the positive semi-definite covariance matrix representing the high-order FC [108].

The matrix Ω effectively captures the correlations among edges (high-order interactions). By assuming the covariance structure decomposes into the Kronecker product $\Omega \otimes \Omega$ (assuming symmetry where row and column covariances are identical), the method avoids the computational intractability of estimating a full $P^2 \times P^2$ covariance matrix, effectively compressing the high-order information back into an ROI $\times$ ROI format [108]. The parameters are estimated via Maximum Likelihood Estimation (MLE) using the following iterative update:

$$\Omega = \frac{1}{KP} \sum_{k=1}^{K} (W^{(k)} - M)\Omega^{-1}(W^{(k)} - M)^T,$$

where $M = \frac{1}{K}\sum_{k=1}^{K} W^{(k)}$. The resulting matrix Ω serves as the final high-order functional connectivity network for downstream analysis [108].

6.7 Hypergraph-Based High-Order FC (H-CHON)

To capture complex interactions among multiple brain regions beyond pairwise connections, **(author?)** [97] proposed a hypergraph-based high-order functional connectivity network (FCN) construction method. This approach leverages the "friend circle" concept to model the intimacy and interaction within groups of ROIs.

Let $V = \{v_1, v_2, \ldots, v_N\}$ denote the set of N brain regions, where each $v_i \in \mathbb{R}^T$ represents the rs-fMRI time series of the i-th ROI. First, a traditional low-order FCN, denoted as $L \in \mathbb{R}^{N \times N}$, is computed using Pearson correlation coefficients between all pairs of ROI time series. For each ROI v_i, a "friend circle" C_i is defined as the set containing v_i and its k nearest neighbors (most strongly correlated regions) based on L:

$$C_i = \{v_j \mid v_j \in \text{top-}k \text{ neighbors of } v_i\} \cup \{v_i\}.$$

Next, to characterize the collective behavior of this group, a representative "friend circle sequence" $S_i \in \mathbb{R}^T$ is generated by averaging the time series of all ROIs within C_i:

$$S_i = \frac{1}{|C_i|} \sum_{v_j \in C_i} v_j.$$

The final high-order functional connectivity matrix $H \in \mathbb{R}^{N \times N}$ is then constructed by calculating the Pearson correlation between these friend circle sequences for every pair of ROIs:

$$H_{ij} = \text{corr}(S_i, S_j), \quad 1 \leq i, j \leq N.$$

This matrix H represents the hypergraph-based high-order network, where each entry H_{ij} reflects the similarity between the functional communities centered at ROI i and ROI j, effectively capturing higher-order interaction patterns that are overlooked by traditional pairwise methods [97].

6.8 Ho-FCN

To overcome the limitations of pairwise correlations in capturing global and complex interaction patterns, **(author?)** [45] proposed a high-order functional connectivity network (Ho-FCN) based on hypergraph theory. Unlike simple graphs where edges connect only two nodes, a hypergraph consists of hyperedges that can connect multiple vertices simultaneously, thereby modeling high-order relationships.

Let $V = \{v_1, v_2, \ldots, v_N\}$ denote the set of N ROIs, where each v_i represents the feature vector (e.g., time series or low-order correlation profile) of the i-th ROI. A hypergraph $G = \{V, E, W\}$ is constructed using a K-nearest neighbor (K-NN) strategy. Specifically, for each ROI v_i, a hyperedge e_i is formed by including v_i and its K nearest neighbors based on the Euclidean distance $d_{ij} = \|v_i - v_j\|_2$. This results in an incidence matrix $H \in \{0, 1\}^{N \times N}$, where entries are defined as:

$$H(v_j, e_i) = \begin{cases} 1 & \text{if } v_j \in e_i, \\ 0 & \text{otherwise.} \end{cases}$$

Here, $H(v_j, e_i)$ indicates whether vertex v_j is part of the hyperedge e_i centered at v_i. To quantify the high-order topology, the normalized hypergraph Laplacian matrix Δ is computed. Let D_v and D_e be

the diagonal matrices representing the degrees of the vertices and hyperedges, respectively, and W be the diagonal matrix of hyperedge weights. The high-order connectivity matrix is defined as:

$$\Delta = I - D_v^{-\frac{1}{2}} H W D_e^{-1} H^T D_v^{-\frac{1}{2}},$$

where I is the identity matrix. This Laplacian matrix Δ serves as the final ROI-by-ROI high-order functional connectivity network, encapsulating the multi-regional interaction information for downstream deep-broad learning analysis [45].

6.9 Multi-Level HOFC

To capture complex interactions among brain regions that go beyond simple pairwise associations, **(author?)** [107] proposed a multi-level high-order functional connectivity (FC) network framework. This approach iteratively constructs higher-level networks based on the "correlation of correlations" strategy.

Let $x_i \in \mathbb{R}^T$ denote the average rs-fMRI time series of the i-th ROI ($i = 1, \ldots, M$), where M is the total number of ROIs. First, a conventional low-order FC network (LON), denoted as $C_{LON} \in \mathbb{R}^{M \times M}$, is constructed by computing the Pearson correlation coefficient between the time series of any pair of ROIs:

$$c_{ij} = \mathrm{corr}(x_i, x_j), \quad 1 \leq i, j \leq M.$$

To derive the first-level high-order network (HON-1), the method defines a connectivity profile $\mathbf{c}_i = [c_{i1}, c_{i2}, \ldots, c_{iM}]^T$ for each ROI i, which represents its functional connectivity pattern with the entire brain. The high-order connectivity between ROI i and ROI j is then quantified by the correlation between their respective connectivity profiles:

$$c_{ij}^{(2)} = \mathrm{corr}(\mathbf{c}_i, \mathbf{c}_j).$$

This results in the first-level high-order matrix $C_{HON-1} = (c_{ij}^{(2)}) \in \mathbb{R}^{M \times M}$. This process can be generalized recursively: the t-th level high-order network C_{HON-t} is derived by correlating the connectivity profiles (rows or columns) extracted from the previous level matrix $C_{HON-(t-1)}$. Each resulting high-order matrix maintains the ROI $\times$ ROI structure while characterizing increasingly abstract functional relationships [107].

6.10 Higher-Order Sparse Graph

To characterize the interactions among multiple brain regions simultaneously, **(author?)** [52] proposed a hyper-connectivity network framework based on sparse representation. Unlike simple graphs where edges connect only two nodes, this method constructs a hyper-graph where a hyper-edge can connect an arbitrary number of nodes, capturing high-order functional relationships.

Let $X = [x_1, \ldots, x_M]^T \in \mathbb{R}^{M \times T}$ denote the fMRI data matrix, where $x_m \in \mathbb{R}^T$ is the regional mean time series of the m-th ROI and M is the total number of ROIs. The method assumes that the activity of one ROI can be reconstructed by a sparse linear combination of the activities of other ROIs. For each ROI x_m, the relationship is modeled as:

$$x_m = A_m \alpha_m + \tau_m,$$

where $A_m = [x_1, \ldots, x_{m-1}, 0, x_{m+1}, \ldots, x_M]$ is the data matrix excluding the m-th ROI, $\alpha_m \in \mathbb{R}^M$ is the weight vector representing the influence of other ROIs, and τ_m is the noise term. To identify the most significant interactions, the weight vector α_m is estimated by solving the following l_1-norm regularized optimization problem (Lasso):

$$\min_{\alpha_m} \|x_m - A_m \alpha_m\|_2^2 + \lambda \|\alpha_m\|_1,$$

where λ is a regularization parameter controlling sparsity. A hyper-edge e_m is then constructed to include the centroid ROI m and all other ROIs corresponding to non-zero elements in α_m. This process forms a hyper-graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, which can be represented by an incidence matrix H or converted into an ROI-by-ROI adjacency matrix A (where A_{ij} counts the shared hyper-edges between regions i and j), effectively mapping high-order interactions into a graph structure for diagnosis [52].

6.11 BHM

To capture high-order functional interactions within a probabilistic framework, **(author?)** [51] proposed a Bayesian High-order Model (BHM) that estimates brain networks by viewing functional connectivity not as fixed parameters, but as random variables generated from a population distribution.

Let $X \in \mathbb{R}^{P\times N}$ denote the rs-fMRI data matrix with P ROIs and N time points. The relationship between ROIs is modeled via a low-order functional connectivity matrix $W \in \mathbb{R}^{P\times P}$, where the signal of one region x_i can be reconstructed by the signals of other regions weighted by W:

$$x_i = \sum_{j\neq i} w_{ij}x_j + \epsilon_i,$$

where ϵ_i is Gaussian noise. To characterize high-order correlations, the connectivity matrix W is assumed to follow a Matrix Variate Normal Distribution (MVND) with mean matrix M and a covariance structure governed by $\Omega \in \mathbb{R}^{P\times P}$:

$$p(W|\Omega) \sim \mathcal{MN}_{P,P}(M, \Omega, \Omega).$$

Here, Ω represents the high-order functional connectivity network, capturing the structural correlations between the connectivity profiles (rows/columns) of the ROIs. The method estimates both the low-order network W and the high-order network Ω by maximizing the posterior probability $p(W, \Omega|X)$ using a Maximum A Posteriori (MAP) estimation framework. The optimization objective combines the data likelihood with the MVND prior:

$$\min_{W,\Omega} \frac{1}{2\sigma^2}\sum_{i=1}^{P} \|x_i - \sum_{j\neq i} w_{ij}x_j\|^2 + \frac{1}{2}\mathrm{tr}(\Omega^{-1}(W-M)\Omega^{-1}(W-M)^T) + P\ln|\Omega|.$$

By solving this, the resulting matrix Ω serves as the ROI-by-ROI high-order adjacency matrix, providing a more robust representation of brain network organization for disease identification [51].

7 End-to-end Graph Construction Static

A growing line of work seeks to learn functional brain graphs directly from ROI-wise BOLD time-series in an end-to-end fashion, letting the downstream objective supervise *how* edges are formed. Unlike two-stage pipelines (compute connectivity first, then learn on a fixed graph), these approaches parameterize a graph generator that maps $X \in \mathbb{R}^{v\times t}$ (ROI $\times$ time) into a subject-specific adjacency $A \in \mathbb{R}^{v\times v}$ (possibly time-varying), and then apply a GNN/graph module for prediction. Below we summarize representative models that satisfy: *input = ROI time-series; output = learned graph.*

7.1 FBNetGen

FBNetGen [53] learns task-aware functional networks from raw BOLD. A temporal encoder (1D-CNN or biGRU) produces ROI embeddings $H_e \in \mathbb{R}^{v\times d}$:

$$H_e = \mathrm{Encoder}(X).$$

A differentiable graph generator then produces a non-negative adjacency via a similarity kernel followed by row-wise normalization, e.g.,

$$A = \mathrm{softmax}\big(H_e H_e^\top\big).$$

A GNN operates on (A, H_e) with layers $H^{(k+1)} = \sigma\big(AH^{(k)}W^{(k)}\big)$ and a graph readout drives the supervised loss. The learned A is task-optimized and interpretable (e.g., edges most predictive of phenotype).

7.2 DABNet: Deep DAG Learning of Effective Connectivity

DABNet [99] targets *effective* (directional) connectivity by learning a directed acyclic graph (DAG) from ROI time-series through a Structural VAR formulation. Let X collect the ROI signals and Z denote lagged copies; DABNet jointly estimates a directed matrix A and lag matrices B by maximizing a likelihood with sparsity and a smooth acyclicity penalty (e.g., $h(A) = \mathrm{tr}\big(e^{A\circ A}\big) - v$). The resulting directed A serves as adjacency for a GNN adapted to signed/typed edges. Training is end-to-end so the DAG generator is shaped by the final task.

7.3 Graph Autoencoders

Graph Autoencoders (GAEs) are advanced methods designed for learning low-dimensional representations (embeddings) of graph-structured data. In the context of brain functional connectivity networks, GAEs leverage graph convolutional networks (GCNs) to preserve both the topological structure of the brain network and the node-specific features, enabling effective classification of brain disorders such as Major Depressive Disorder (MDD).

To construct functional connectivity networks, we begin with a raw fMRI time-series matrix of shape $t \times n$, where t is the number of time points and n is the number of regions of interest (ROIs). Each subject's brain network is modeled as an undirected graph $G = \{V, E\}$, where:

- $V = \{v_1, v_2, \ldots, v_n\}$ represents the set of n ROIs (nodes),
- E represents the set of edges encoding functional connectivity between ROIs.

The graph structure G is represented by its adjacency matrix $A \in \mathbb{R}^{n \times n}$, and the node-specific features are stored in a matrix $X \in \mathbb{R}^{n \times d}$, where d is the feature dimension.

The GAE architecture consists of two primary components:

1. **Graph Encoder**: A GCN-based encoder maps the input graph structure A and node features X to a low-dimensional embedding $Z \in \mathbb{R}^{n \times k}$, where $k \ll n$. The GCN encoder applies spectral graph convolutions to aggregate neighborhood information:

$$Z^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} Z^{(l)} W^{(l)}\right), \tag{39}$$

   where:

   - $\tilde{A} = A + I_n$ is the adjacency matrix with self-connections,
   - $\tilde{D}$ is the degree matrix of $\tilde{A}$,
   - $Z^{(l)}$ is the input feature matrix at layer l, with $Z^{(0)} = X$,
   - $W^{(l)}$ are the trainable weights, and $\sigma(\cdot)$ is the activation function.

2. **Graph Decoder**: The decoder reconstructs the graph adjacency matrix A from the learned embeddings Z. This is typically done using an inner-product decoder:

$$\hat{A} = \sigma(ZZ^T), \tag{40}$$

   where $\hat{A}$ is the reconstructed adjacency matrix, and $\sigma(\cdot)$ is the logistic sigmoid function.

The GAE framework is trained by minimizing the reconstruction loss, which measures the discrepancy between the original and reconstructed adjacency matrices:

$$L = -\sum_{i=1}^{n} \sum_{j=1}^{n} \left[A_{ij} \log \hat{A}_{ij} + (1 - A_{ij}) \log(1 - \hat{A}_{ij})\right]. \tag{41}$$

The learned node embeddings Z capture both the graph topology and node feature information. To obtain a graph-level representation for downstream tasks, a readout operation (e.g., mean, sum, or max pooling) is applied to the node embeddings:

$$z_G = \text{readout}(Z) \in \mathbb{R}^k. \tag{42}$$

The resulting graph embeddings z_G are then used as inputs to a fully-connected neural network (FCNN) for classification tasks.

7.4 BrainGNN

[63]

7.5 DBGSL: Dynamic Brain Graph Structure Learning

DBGSL [16] learns *time-varying* graphs directly from sequences. A spatio-temporal encoder produces ROI features $h_i^{(t)}$ and a differentiable attention-based graph constructor yields a dynamic adjacency $A^{(t)}$ with entries derived from pairwise compatibilities (e.g., dot-product or MLP). Message passing proceeds on $\{A^{(t)}\}_{t=1}^{T}$ (via temporal graph convolutions or recurrent units), and the whole model is trained to predict subject labels. Thus, DBGSL jointly learns *when* and *how* ROIs interact.

7.6 BrainDGT: Dynamic Graph Transformer

BrainDGT [82] deconvolves the hemodynamic response and uses a transformer to infer dynamic graphs within functional modules, producing a sequence $\{A^{(t)}\}$ from ROI time-series. Local (module-level) attention forms fine-grained edges while a global fusion captures cross-module interactions. The attention weights function as learned, time-resolved connectivities optimized for diagnosis.

8 Temporal Stastical Estimation

8.1 Fixed Sliding Window

The fixed sliding window (FSW) approach is one of the earliest and most widely used techniques for modeling dynamic functional connectivity in resting-state fMRI, having been popularized through large-scale investigations of connectivity fluctuations [2] and subsequently evaluated in methodological analyses [79]. In this method, the BOLD time series $\mathbf{X} \in \mathbb{R}^{t\times n}$ is divided into overlapping segments of fixed duration. Given a window length w and step size s, a sequence of m windows $\{\mathbf{X}^{(k)}\}_{k=1}^{m}$ is generated as

$$\mathbf{X}^{(k)} = \mathbf{X}[(k-1)s+1 : (k-1)s+w].$$

Within each window, a connectivity estimator $f(\cdot,\cdot)$ is applied to obtain a time-resolved adjacency matrix $A^{(k)} \in \mathbb{R}^{n\times n}$, where

$$A_{ij}^{(k)} = f\big(x_i^{(k)}, x_j^{(k)}\big).$$

In practice, a broad family of estimators has been embedded in the FSW framework. For example, Savva et al. systematically compared ten windowed metrics, including: (i) Pearson correlation[79], (ii) partial Pearson correlation[79], (iii) sparse inverse covariance estimates[79], (iv) Spearman correlation[79], (v) partial Spearman correlation[79], (vi) Kendall's τ[79], (vii) mutual information[79], (viii) variation of information[79], (ix) symmetrized Kullback–Leibler divergence[79], and (x) the multiplication of temporal derivatives (MTD)[79]. These variants illustrate that FSW is agnostic to the specific choice of connectivity metric and can accommodate linear, rank-based, information-theoretic, and derivative-based measures within a unified windowed framework.

The resulting graph sequence

$$\mathcal{A} = \{A^{(1)}, A^{(2)}, \ldots, A^{(m)}\}$$

captures the temporal evolution of ROI-to-ROI interactions across the scan. The FSW method is conceptually simple, computationally accessible, and compatible with standard correlation-based estimators, which has contributed to its extensive use in identifying recurring connectivity configurations and studying whole-brain temporal dynamics. However, its reliance on manually selected window parameters has motivated the development of tapered, adaptive, and model-based variants that aim to improve robustness and interpretability.

8.2 Tapered Sliding Window

The tapered sliding window is a commonly used refinement of the fixed-window approach, introduced to mitigate the sensitivity of dynamic correlation estimates to abrupt boundary effects and influential outliers [59]. Rather than assigning uniform weights to all observations within a window, this method applies a smoothing kernel so that samples gradually enter and exit the window as it moves across time. Let $h(\tau)$ denote a tapering function, typically a Gaussian kernel, defined over the window of length w. For the k-th segment $\mathbf{X}^{(k)} \in \mathbb{R}^{w\times n}$, the tapered time series for ROI i is given by $x_i^{(k)}(\tau)h(\tau)$, where $\tau = 1, \ldots, w$. A tapered correlation matrix $A^{(k)} \in \mathbb{R}^{n\times n}$ is computed by replacing the raw windowed signals with their tapered versions:

$$A_{ij}^{(k)} = f\big(x_i^{(k)}(\tau)h(\tau),\, x_j^{(k)}(\tau)h(\tau)\big).$$

The resulting sequence of matrices $\mathcal{A} = \{A^{(1)}, A^{(2)}, \ldots, A^{(m)}\}$ provides a smoothed estimate of time-varying interactions that is less susceptible to sharp transitions introduced by the window edges. By emphasizing central observations and down-weighting points near the boundaries, the tapered sliding window reduces spurious fluctuations and yields a more stable characterization of dynamic connectivity patterns while preserving compatibility with standard correlation-based estimators.

8.3 Adaptive Sliding Window

Adaptive sliding window methods extend the conventional fixed-window framework by allowing the window width to vary across time in response to local signal characteristics. This approach is motivated by the observation that dynamic correlation estimates require wider windows in slowly varying regions and narrower windows when the underlying connectivity changes rapidly. A representative method is the adaptive windowed time-varying correlation coefficient estimator introduced in [105], which formulates window selection as a data-driven bandwidth optimization problem within a kernel-smoothing framework. Let $h(n)$ denote the locally selected bandwidth at time n. The adaptive estimator replaces the fixed window with a kernel-weighted segment whose width is determined by $h(n)$, computing the time-varying covariance as

$$\mathrm{Cov}(n;h) = \sum_{m=-h(n)}^{h(n)} K_{h(n)}(m)\, x(n-m)x^{\top}(n-m),$$

where $K_{h(n)}$ is a symmetric kernel. The corresponding time-varying correlation matrix is obtained by normalizing the entries of $\mathrm{Cov}(n;h(n))$ at each time point. The local bandwidth $h(n)$ is selected via a plug-in rule that minimizes the mean squared error of the covariance estimator, using iterative refinement of kernel-based estimates of local derivatives. This yields an adaptive sequence of connectivity matrices $\mathcal{A} = \{A(1), A(2), \ldots, A(t)\}$ in which each estimate is computed with a window size tailored to the local dynamics of the BOLD signals. By accommodating spatially and temporally heterogeneous fluctuations, the adaptive sliding window mitigates the bias–variance tradeoff inherent to fixed-size windows and provides a more responsive characterization of transient functional interactions while remaining compatible with correlation-based estimators.

8.4 Exponential Weighted Moving Average (EWMA)

The Exponential Weighted Moving Average (EWMA) method provides a parametric and smoothly adaptive alternative to fixed sliding-window estimators for dynamic functional connectivity [59]. Instead of assigning uniform weights within a window and zero weight outside it, EWMA applies exponentially decaying weights to past observations, allowing older samples to gradually diminish in influence while more recent observations dominate the covariance estimate.

EWMA models the time-varying covariance recursively as

$$\Sigma_t = (1-\lambda)\, e_{t-1}e_{t-1}^{\top} + \lambda\, \Sigma_{t-1},$$

where $\lambda \in (0,1)$ controls the rate of exponential decay and e_{t-1} denotes the demeaned observation at time $t-1$. Expanding this recursion yields the individual variance and covariance updates,

$$\sigma_{i,t}^2 = (1-\lambda)\, y_{i,t-1}^2 + \lambda\, \sigma_{i,t-1}^2,$$

$$\sigma_{ij,t} = (1-\lambda)\, e_{i,t-1}e_{j,t-1} + \lambda\, \sigma_{ij,t-1}.$$

Because λ can be estimated through maximum likelihood rather than chosen arbitrarily, EWMA effectively induces a data-driven adaptive window length that better tracks gradual or abrupt changes in the signal. As a result, it provides a smoother and more statistically principled estimate of dynamic covariance and correlation trajectories than classical sliding-window approaches.

8.5 Hilbert-based Instantaneous Phase Synchronization

Hilbert-based instantaneous phase synchronization defines dynamic connectivity by tracking the moment-to-moment alignment of narrowband BOLD signal phases. Following [39], each ROI time series $x_i(t)$ is first band-limited to a low-frequency range (e.g., 0.04–0.07 Hz) to ensure that an instantaneous phase is meaningful under the narrowband assumption. The analytic signal of $x_i(t)$ is then obtained via the Hilbert transform,

$$x_{a,i}(t) = x_i(t) + j\, \mathcal{H}\big[x_i(t)\big],$$

where $\mathcal{H}[\cdot]$ denotes the Hilbert transform and j is the imaginary unit [39]. Writing $x_{a,i}(t)$ in polar form,

$$x_{a,i}(t) = a_i(t)\, e^{j\phi_i(t)},$$

yields a time-varying amplitude $a_i(t)$ and instantaneous phase $\phi_i(t)$ for each ROI. Dynamic connectivity is then quantified from the phase differences between regions. A simple instantaneous phase-synchrony adjacency can be defined at each time point as

$$A_{ij}(t) = \cos\big(\phi_i(t) - \phi_j(t)\big),$$

which yields a sequence of phase-synchrony graphs $\{A(t)\}_{t=1}^{T}$ without requiring an explicit sliding window. Within this framework, [39] introduced several group-level metrics that aggregate phase information across subjects and brain regions, described below.

8.5.1 Seed-based Phase Synchronization (SBPS)

Seed-based phase synchronization (SBPS) provides a group-level measure of dynamic functional connectivity between two regions, while accounting for individual differences in their phase relationship [39]. Let $\phi_{s,1}(t)$ and $\phi_{s,2}(t)$ denote the instantaneous phases for subject s at two ROIs (or voxels). The subject-specific phase difference is

$$\Delta\phi_s(t) = \phi_{s,1}(t) - \phi_{s,2}(t).$$

The seed-based phase coefficient (SBPC) is defined as the circular mean of these phase differences over S subjects,

$$\mathrm{SBPC}(t) = \frac{1}{S}\sum_{s=1}^{S} \exp\big(j\,\Delta\phi_s(t)\big).$$

The complex number $\mathrm{SBPC}(t)$ encodes both the mean phase difference and its dispersion across subjects. Let

$$\mu_{\Delta\phi}(t) = \arg\left(\sum_{s=1}^{S} \exp\big(j\,\Delta\phi_s(t)\big)\right)$$

denote the mean angle of the phase differences. The seed-based phase synchronization (SBPS) is then given by

$$\mathrm{SBPS}(t) = \cos\big(\mu_{\Delta\phi}(t)\big)\,\big\|\mathrm{SBPC}(t)\big\|,$$

which takes values in $[-1, 1]$ and reflects both the strength and sign (in-phase vs. anti-phase) of group-level synchrony between the two regions [39]. This yields a time-resolved, group-level edge weight between the seed and target ROI, forming a dynamic connectivity trajectory for that pair.

8.5.2 Intersubject Phase Synchronization (IPS)

Intersubject phase synchronization (IPS) measures the reliability of activity at a given location across subjects, analogous to intersubject correlation (ISC) but expressed in terms of instantaneous phase [39]. For a voxel or ROI r, let $\phi_{s,r}(t)$ be the phase for subject s at time t. The intersubject phase coefficient (IPC) is defined as

$$\mathrm{IPC}_r(t) = \frac{1}{S}\sum_{s=1}^{S} \exp\big(j\,\phi_{s,r}(t)\big).$$

The IPS time course at location r is then the length of this complex vector,

$$\mathrm{IPS}_r(t) = \big\|\mathrm{IPC}_r(t)\big\|,$$

which lies in $[0, 1]$. High IPS indicates that subjects share a similar phase at that time point (strong across-subject reliability), whereas low IPS indicates phase dispersion [39]. Because IPS is defined at each TR without windowing, it provides a high-temporal-resolution measure of group-level synchronization and serves as a phase-based alternative to sliding-window ISC.

8.5.3 Intersubject Seed-based Phase Synchronization (ISBPS)

Intersubject seed-based phase synchronization (ISBPS) extends IPS from voxel-wise activity to connectivity, quantifying the reliability of ROI-to-ROI phase relationships across subjects [39]. Let $\phi_{s,r}(t)$ denote the phase of subject s at region r, and consider a set of R regions (e.g., seed and target ROIs). The intersubject seed-based phase coefficient (ISBPC) is defined as

$$\mathrm{ISBPC}(t) = \frac{1}{SR}\sum_{s=1}^{S}\sum_{r=1}^{R} \exp\big(j\,\phi_{s,r}(t)\big),$$

and the corresponding ISBPS is given by its magnitude,

$$\mathrm{ISBPS}(t) = \big\|\mathrm{ISBPC}(t)\big\|.$$

In the pairwise case (two regions per subject), ISBPS takes values in $[0, 1]$ and is high only when the phase relationship between the two regions is consistent both within and across subjects. Thus, ISBPS can be interpreted as a dynamic measure of connectivity-level intersubject reliability, complementing SBPS (which summarizes group-level connectivity for a specific seed–target pair) and IPS (which summarizes group-level activity at single locations) [39].

8.6 Phase Locking Value (PLV)

Phase locking value (PLV) is a classical measure of phase synchronization that has been widely used in EEG/MEG and can be adapted for dynamic fMRI connectivity once narrowband phases are defined [56]. Given instantaneous phases $\phi_i(t)$ and $\phi_j(t)$ for two ROIs within a time window $\mathcal{W}$ of length $|\mathcal{W}|$, the PLV is defined as

$$\mathrm{PLV}_{ij} = \left| \frac{1}{|\mathcal{W}|} \sum_{t \in \mathcal{W}} \exp\big(j\,(\phi_i(t) - \phi_j(t))\big) \right|.$$

PLV takes values in $[0, 1]$: it equals 1 when the phase difference $\phi_i(t) - \phi_j(t)$ is constant over time (perfect phase locking), and approaches 0 when phase differences are uniformly distributed. Dynamic PLV-based connectivity can be obtained by computing PLV in a sliding window, yielding a sequence of adjacency matrices $\{A^{(k)}\}_{k=1}^{m}$ whose entries are PLV values for each ROI pair and window k. In the limiting case where $|\mathcal{W}| = 1$, PLV reduces to an instantaneous measure analogous to the Hilbert-based phase synchrony described above, but the windowed formulation is often preferred to stabilize estimates in noisy data.

8.7 Wavelet Transform Coherence (WTC) dFC

Wavelet transform coherence extends sliding-window dynamic connectivity into the time–frequency domain by quantifying the frequency-specific synchrony between pairs of BOLD signals. This method was introduced for fMRI connectivity dynamics in [17], which demonstrated that windowed spectral measures can reveal transient coupling not visible in static correlations. Given ROI time series $\{x_i(t)\}_{i=1}^{n}$, the continuous wavelet transform $W_x(t, f)$ provides a joint time–frequency representation of each signal. The wavelet coherence between ROI i and ROI j is computed as

$$C_{ij}(t, f) = \frac{\Big|S\Big(W_{x_i}(t, f)\, W^*_{x_j}(t, f)\Big)\Big|^2}{S(|W_{x_i}(t, f)|^2)\; S\big(|W_{x_j}(t, f)|^2\big)},$$

where $S(\cdot)$ denotes local smoothing in time and frequency. To obtain a dynamic functional connectivity sequence, the coherence values are either averaged across a frequency band of interest or summarized through a weighted integration across frequencies. This yields, for each time point or window, a coherence-based adjacency matrix $A^{(k)} \in \mathbb{R}^{n \times n}$ capturing frequency-resolved coupling. The full output is a time-indexed series $\mathcal{A} = \{A^{(1)}, A^{(2)}, \ldots, A^{(m)}\}$ providing a dynamic characterization of connectivity in the joint time–frequency domain. By incorporating spectral selectivity, WTC dFC captures transient oscillatory interactions that may be obscured when using purely time-domain correlations.

8.8 Dynamic Conditional Correlation (DCC) dFC

Dynamic Conditional Correlation models provide a likelihood-based framework for estimating time-resolved functional connectivity without relying on fixed sliding windows. Originally developed for multivariate volatility modeling, DCC has been evaluated as a candidate approach for fMRI dynamic connectivity in [59]. Given an ROI time series $x(t) \in \mathbb{R}^n$, the method first fits a univariate GARCH(1,1) model to each ROI to estimate time-varying conditional variances,

$$h_{i,t} = \omega_i + \alpha_i \epsilon^2_{i,t-1} + \beta_i h_{i,t-1},$$

and obtains standardized residuals $z_{i,t} = \epsilon_{i,t}/\sqrt{h_{i,t}}$. Dynamic correlations are then computed by updating an intermediate matrix Q_t according to

$$Q_t = (1 - a - b)\bar{Q} + a\, z_{t-1} z_{t-1}^{\top} + b\, Q_{t-1},$$

where $\bar{Q}$ is the empirical covariance of standardized residuals and a, b control the temporal smoothing. Normalizing Q_t yields the dynamic correlation matrix,

$$R_t = \mathrm{diag}(Q_t)^{-1/2}\, Q_t\, \mathrm{diag}(Q_t)^{-1/2}.$$

The resulting sequence $\{R_t\}_{t=1}^{T}$ provides a fully time-resolved set of ROI–ROI connectivity matrices, with each R_t representing the conditional correlation structure at time t. DCC thus produces an ROI×ROI×T dynamic connectivity series with higher temporal resolution than sliding-window methods, while imposing a parametric temporal structure that stabilizes correlation estimates across successive TRs.

9 Temporal Latent Modeling

9.1 HMM

Hidden Markov models (HMMs) offer a window-free and probabilistic framework for estimating dynamic functional connectivity by assuming that the fMRI time series arise from a sequence of latent brain states. Following [92], the BOLD signals are represented as multivariate observations $\mathbf{x}(t) \in \mathbb{R}^n$ where each dimension corresponds to an ROI. The HMM posits that at each time point t, the data are generated from one of K hidden states, and each state $z(t) \in \{1, \ldots, K\}$ is characterized by a state-specific connectivity model. In the Gaussian HMM formulation, state k is parametrized by a covariance matrix $\Sigma_k \in \mathbb{R}^{n\times n}$ that describes the instantaneous functional interactions among ROIs when the system occupies that state. The probability of being in each state evolves according to a first-order Markov chain, allowing rapid transitions between distinct connectivity configurations.

Once the HMM is trained, the state posterior probabilities yield a time-resolved state sequence, and each time point is associated with the covariance matrix of its most likely state. This produces a piecewise constant dynamic connectivity series $\mathcal{A} = \{A(1), A(2), \ldots, A(T)\}$ where $A(t) = \Sigma_{z(t)}$. Although updates occur only at state transitions, the resulting sequence retains full temporal indexing and provides a principled estimate of time-varying ROI-to-ROI interactions. Compared with sliding-window approaches, HMM-based dFC avoids window-length selection, supports higher temporal precision, and captures transient, recurrent connectivity states that reflect meaningful network reconfigurations across time.

9.2 Kalman Filter Based Time-Varying Graphical Models

State-space formulations provide a principled framework for estimating dynamic brain connectivity by allowing the parameters of a multivariate autoregressive (MVAR) model to evolve over time. Following [69], the BOLD signals from n ROIs are modeled using a time-varying MVAR(p) process,

$$x(t) = \sum_{k=1}^{p} A_k(t)x(t-k) + e(t),$$

where $A_k(t) \in \mathbb{R}^{n\times n}$ denotes the time-dependent connectivity matrix at lag k, and $e(t)$ is zero-mean Gaussian noise. The evolution of connectivity is captured by a random-walk state equation applied to the vectorized parameter set $\theta(t) = \mathrm{vec}([A_1(t), \ldots, A_p(t)])$,

$$\theta(t) = \theta(t-1) + w(t), \qquad w(t) \sim \mathcal{N}(0, Q),$$

which assumes that changes in connectivity occur smoothly across time. These equations define a linear Gaussian state-space model, enabling sequential estimation of $\theta(t)$ via Kalman filtering. At each time step, the filter performs a prediction–correction update based on the mismatch between the observed signal and its one-step MVAR prediction,

$$\theta_{t|t} = \theta_{t|t-1} + K_t\big(x(t) - \hat{x}(t|\theta_{t|t-1})\big),$$

where K_t is the Kalman gain derived from the state and observation covariances.

The resulting estimates of $A_k(t)$ at every time index provide a fully time-resolved sequence of connectivity matrices $A(t)$, representing a dynamic graphical model updated continuously without the need for sliding windows. This approach yields high temporal resolution and enforces temporal smoothness through the state dynamics, making it well suited for capturing rapid but structured fluctuations in functional interactions.

9.3 Time-Varying Graphical Lasso

Time-varying graphical models aim to recover a sequence of evolving connectivity graphs by estimating a separate precision matrix at each time index while enforcing temporal smoothness across consecutive estimates. The framework introduced in [43] formulates this problem as a regularized optimization over a set of precision matrices $\{\Theta_1, \ldots, \Theta_T\}$, where each $\Theta_t \in \mathbb{R}^{n\times n}$ encodes the conditional dependencies among ROIs at time t. Given empirical covariance estimates $\{S_t\}$ derived from local windows or sufficient statistics, the method solves

$$\min_{\Theta_1,\ldots,\Theta_T} \sum_{t=1}^{T} \big(-\log\det\Theta_t + \operatorname{tr}(S_t\Theta_t) + \lambda\|\Theta_t\|_1 \big) \; + \; \beta \sum_{t=2}^{T} \|\Theta_t - \Theta_{t-1}\|_F,$$

where the ℓ_1 penalty promotes sparse connectivity while the fused penalty encourages adjacent precision matrices to remain similar unless supported by the data. This yields a sequence of dynamic graphs that can change smoothly or abruptly depending on the temporal regularization parameter β.

By jointly estimating $\{\Theta_t\}$, the model produces a time-resolved series of ROI–ROI connectivity structures without requiring explicit sliding-window correlation matrices. Each Θ_t represents a graphical model capturing conditional independence relations among ROIs at time t, and the temporal coupling term ensures coherent evolution of network topology across the scan. The resulting output is a fully dynamic graphical representation of functional connectivity, suitable for tracking gradual connectivity fluctuations as well as transient reconfigurations in large-scale brain networks.

9.4 Kalman Filter State-Space TVAR dFC

Time-varying autoregressive models provide a flexible dynamic graphical framework by allowing directed functional interactions among ROIs to evolve continuously over time. Following the state–space estimation principles described in [68], the fMRI signal from n ROIs is modeled using a time-varying vector autoregressive process of order p,

$$y(t) = \sum_{k=1}^{p} A_k(t)\, y(t-k) + \varepsilon(t),$$

where $A_k(t) \in \mathbb{R}^{n\times n}$ encodes the directed connectivity at lag k, and $\varepsilon(t)$ is Gaussian noise. To capture temporal variation in connectivity, the AR parameters are embedded in a linear state–space system by stacking them into a state vector $\theta(t) = \operatorname{vec}\big([A_1(t), \ldots, A_p(t)]\big)$ that evolves according to a random walk,

$$\theta(t) = \theta(t-1) + w(t), \qquad w(t) \sim \mathcal{N}(0, Q).$$

This specification constrains changes in connectivity to be smooth but data-driven across time. Given observations $y(t)$, the Kalman filter updates $\theta(t)$ sequentially through a prediction step based on the state model and a correction step based on the mismatch between the observed signal and its one-step prediction,

$$\theta_{t|t} = \theta_{t|t-1} + K_t\big(y(t) - \hat{y}(t|\theta_{t|t-1})\big),$$

with K_t denoting the Kalman gain. After filtering, the estimated matrices $\{A_k(t)\}$ provide a fully time-resolved sequence of directed connectivity patterns. The resulting output is an ROI–ROI dynamic network $A(t)$ defined at each time index, forming a continuous series of graphical models that capture gradual and transient fluctuations in effective functional interactions.

9.5 Time-Varying VAR Effective Connectivity States

Time-varying vector autoregressive (TV-VAR) modeling provides a directed dynamic connectivity framework by allowing effective interactions among ROIs to evolve smoothly over time. Following the formulation in [78], the fMRI BOLD signal from n ROIs is expressed using a time-varying VAR(p) model,

$$m_t = \sum_{k=1}^{p} A_k(t)\, m_{t-k} + v_t,$$

where $A_k(t) \in \mathbb{R}^{n\times n}$ denotes the directed connectivity matrix at lag k, and v_t is Gaussian observation noise. The VAR coefficients are then placed in a linear state–space model by stacking them into a state vector $\theta_t = \operatorname{vec}\big([A_1(t), \ldots, A_p(t)]\big)$ that evolves according to a random-walk prior,

$$\theta_t = \theta_{t-1} + w_t, \qquad w_t \sim \mathcal{N}(0, Q),$$

while the observation equation takes the form

$$m_t = C_t\,\theta_t + v_t,$$

with C_t constructed from lagged BOLD signals. Estimation proceeds using a forward Kalman filter followed by a Rauch–Tung–Striebel smoother to obtain optimal time-resolved estimates of θ_t, and an expectation–maximization procedure is used to update the noise covariances Q and Σ.

The resulting sequence of coefficient vectors $\{\theta_t\}$ encodes a directed ROI–ROI connectivity matrix at each time point, yielding a fully time-resolved effective connectivity series. To identify recurring connectivity patterns, the method clusters the smoothed parameter trajectories using k-means, producing a set of dynamic effective connectivity states and associated state transition boundaries. This framework, therefore, combines continuous-time estimation of directed interactions with discrete state identification, offering a complementary perspective to windowed correlation, HMM-based, or precision-matrix-based dynamic connectivity approaches.

9.6 Deep Bayesian Generative Dynamic Graph Model (DBGDGM)

Deep generative modeling provides a principled approach for decomposing dynamic functional connectivity into temporally evolving latent structures. The DBGDGM framework introduced in [15] assumes that each dynamic connectivity snapshot $A^{(t)} \in \mathbb{R}^{n\times n}$ is generated from a set of hierarchical latent variables, including node-specific embeddings $z_i^{(t)}$, a global community-level embedding $c^{(t)}$, and a subject-level embedding u. The probability of an edge (i,j) at time t is modeled through a logistic generative function of the latent variables,

$$A_{ij}^{(t)} \sim p\big(A_{ij} \mid z_i^{(t)}, z_j^{(t)}, c^{(t)}, u\big),$$

with temporal dynamics imposed through a Gaussian Markov transition on the node embeddings,

$$z^{(t)} \sim \mathcal{N}\big(\mu(z^{(t-1)}), \Sigma\big),$$

ensuring smooth evolution of latent representations across time. The model is trained by maximizing a variational evidence lower bound, jointly learning the latent trajectories and the generative parameters in an entirely unsupervised manner, without reliance on behavioral or clinical labels.

Given an observed sequence of dynamic connectivity matrices $\{A^{(t)}\}_{t=1}^{T}$, DBGDGM produces reconstructed graphs

$$\hat{A}^{(t)} = f\big(z^{(t)}, c^{(t)}, u\big),$$

forming an ROI–ROI–T dynamic adjacency series that captures time-varying network structure through a low-dimensional latent decomposition. Because the method operates on precomputed graph snapshots rather than raw fMRI signals, it provides a generative characterization of dynamic connectivity rather than end-to-end graph construction, expressing moment-to-moment changes in functional interactions through interpretable latent embeddings and their temporal transitions.

9.7 Change-Point Detection dFC

Change-point based dynamic connectivity methods aim to detect abrupt transitions in the covariance structure of ROI time series and estimate a separate connectivity graph for each stationary segment. The framework of [23] begins by computing sliding-window covariance matrices from the fMRI signals. For an ROI time series $X(t) \in \mathbb{R}^n$, a window of length L centered at time t yields the sample covariance

$$P_t = \frac{1}{L-1} \sum_{\tau=t-L/2}^{t+L/2} \left(X(\tau)-\mu_t\right)\left(X(\tau)-\mu_t\right)^{\top},$$

where μ_t is the local window mean. Each covariance matrix P_t lies on the manifold of symmetric positive-definite matrices, and distances between pairs are computed using the affine-invariant Riemannian metric

$$d(P_i, P_j) = \left\| \log\!\left(P_i^{-1/2} P_j P_i^{-1/2}\right) \right\|_F .$$

Constructing a minimal spanning tree (MST) on all covariance matrices, the method evaluates each candidate boundary τ by counting the number of MST edges that connect points belonging to opposite sides of the partition. Let $R(\tau)$ denote the number of such cross edges; the normalized statistic

$$Z(\tau) = \frac{R(\tau) - \mathbb{E}[R(\tau)]}{\sqrt{\mathrm{Var}[R(\tau)]}}$$

is compared against a bootstrap null distribution to determine whether a significant change point occurs at τ. This yields a set of non-overlapping stationary intervals. For each interval, an ROI-level connectivity graph is estimated by applying graphical Lasso to the Karcher mean covariance $\bar{P}$ of matrices within the segment:

$$\widehat{\Theta} = \arg\min_{\Theta \succeq 0} \operatorname{tr}(\bar{P}\Theta) - \log\det\Theta + \rho \sum_{i \neq j} |\Theta_{ij}|.$$

The final output is a collection of piecewise-constant ROI–ROI networks $\{\widehat{\Theta}_1, \ldots, \widehat{\Theta}_K\}$, each representing the connectivity pattern of a stationary temporal segment.

9.8 Structurally-Regularized Dynamic Dictionary Learning (sr-DDL)

Dynamic dictionary learning provides a low-rank decomposition framework for representing time-varying functional connectivity as a combination of interpretable basis networks. The structurally-regularized DDL method introduced in [27] factorizes a sequence of windowed connectivity matrices $\{\Gamma^t\}_{t=1}^T$ into a set of shared spatial bases and time-varying coefficients. Each dynamic matrix is approximated as

$$\Gamma^t \approx B \operatorname{diag}(c^t)\, B^\top,$$

where $B \in \mathbb{R}^{n \times K}$ contains K latent basis networks shared across all time points, and $c^t \in \mathbb{R}^K$ provides a non-negative weight vector indicating the contribution of each basis at time t. Structural information from diffusion MRI is incorporated as a regularizer that constrains the learned bases to be consistent with underlying white-matter architecture. The optimization jointly estimates B and $\{c^t\}$ under low-rank and structural priors, yielding a compact decomposition of the dynamic connectivity sequence.

This formulation produces a reconstructed time-resolved adjacency series

$$\hat{\Gamma}^t = B \operatorname{diag}(c^t)\, B^\top, \quad t = 1, \ldots, T,$$

which serves as the dynamic functional connectivity output. Because the decomposition is entirely unsupervised and does not use clinical or behavioral labels, the learned bases capture intrinsic temporal patterns of connectivity rather than task- or diagnosis-driven effects. The model therefore provides an interpretable generative representation of dynamic connectivity, expressing each ROI–ROI matrix as a structured mixture of a small number of neurobiologically constrained basis networks.

10 End-to-end Graph Construction Dynamic

10.1 Dynamic Directed Connectivity via DICE

Deep learning models enable end-to-end estimation of dynamic connectivity by learning temporal representations directly from the ROI time series and generating a directed adjacency matrix at each time point. The DICE framework introduced in [64] adopts this strategy by embedding each ROI signal into a time-dependent latent representation through a bidirectional LSTM. Given the fMRI sequence $x_i(1{:}T)$ for ROI i, the model computes

$$h_i(t) = \mathrm{BiLSTM}\big(x_i(1{:}t)\big),$$

providing a temporally informed feature vector for each node at each time index. Dynamic directed connectivity is then obtained by applying a self-attention mechanism across the ROI embeddings. For every pair of regions (i, j), the time-varying directed edge weight is computed as

$$W_t(i,j) = \mathrm{softmax}_j\big(q_i(t)^\top k_j(t)\big),$$

where $q_i(t)$ and $k_j(t)$ are learned query and key projections of $h_i(t)$ and $h_j(t)$. This produces a full directed adjacency matrix $W_t \in \mathbb{R}^{n \times n}$ at each time point t, yielding the dynamic graph sequence $\{W_t\}_{t=1}^T$ that characterizes instantaneous interactions among ROIs.

To enhance interpretability at the sequence level, DICE further aggregates the dynamic adjacency matrices using a temporal attention mechanism that learns weights α_t reflecting the relative contribution of each time point. Although this aggregation is used for downstream tasks, the primary output of the model is the set of directed ROI–ROI connectivity matrices computed at every TR. The method therefore provides an end-to-end, model-based construction of dynamic directed graphs without requiring sliding windows, parametric state models, or explicit covariance estimation.

10.2 Dynamic Effective Connectivity via CAST

The CAST framework provides an end-to-end procedure for estimating time-resolved effective connectivity directly from fMRI time series. Building on the generalized psychophysiological interaction model, CAST computes directed connectivity for each ROI pair by fitting, at every window, a multivariate regression of the form

$$y_j(t) = \beta_0 + \beta_1 \,\mathrm{Psy}(t) + \beta_2 \, y_i(t) + \beta_3 \big(\mathrm{Psy}(t) \, y_i(t)\big) + \varepsilon(t),$$

where $y_i(t)$ and $y_j(t)$ denote the BOLD signals of ROIs i and j, $\mathrm{Psy}(t)$ encodes task condition, and the interaction coefficient β_3 represents the directed effective influence from i to j. To capture temporal variation, the model applies overlapping windows across the scan, estimating β_3 for each window and forming an initial sequence of effective connectivity matrices.

To ensure statistical validity and temporal comparability, CAST imposes a global null hypothesis test on each edge trajectory across all windows, retaining only those connections whose dynamic profiles significantly deviate from zero. Surviving edges are further normalized using an ℓ_2 scaling within each window to stabilize magnitude fluctuations over time. The resulting output is a directed ROI–ROI dynamic graph sequence $\{E^{(t)}\}_{t=1}^{T}$, where each $E^{(t)}$ is an effective connectivity matrix reflecting task-modulated interactions. This yields an interpretable, time-resolved effective connectivity representation that integrates task design, directed influence modeling, statistical sparsification, and temporal normalization in a fully data-driven analytic pipeline [18].

10.3 Class-Aware HMM for Joint dFC Estimation and Classification

The Class-Aware Hidden Markov Model proposed in [44] aims to integrate dynamic functional connectivity estimation and subject-level classification into a unified probabilistic framework. The key motivation is that conventional HMM-based dynamic FC pipelines rely on a two-step procedure—first estimating latent states and state-specific connectivity patterns, and then extracting secondary features for classification—which may introduce inefficiencies and reduce sensitivity. To address this limitation, the authors propose a supervised extension of HMMs based on Hidden Conditional Random Fields (HCRF), allowing the latent state sequence to be optimized jointly with the class-discriminative objective.

According to the available abstract, the method retains the standard HMM assumption that each latent state corresponds to a functional connectivity pattern, while HCRF introduces a discriminative mapping from the full latent state path to the class label. This enables a one-step estimation procedure where dynamic states and their associated connectivity patterns are directly shaped by the classification task. However, the full manuscript is not publicly accessible, and the detailed mathematical formulation, estimation strategy, and the precise nature of the connectivity outputs cannot be verified at this time. As such, the method is included here for completeness, but its exact implementation and compatibility with other dynamic FC frameworks remain to be confirmed.

11 Experiments

11.1 Datasets

11.1.1 ADNI

The Alzheimer's Disease Neuroimaging Initiative (ADNI) is a multi-site, longitudinal cohort designed to accelerate biomarker discovery and clinical trial readiness for Alzheimer's disease (AD) across the spectrum from cognitively normal aging through mild cognitive impairment (MCI) to dementia [7, 50, 70, 95]. While ADNI's core modalities include structural MRI, PET, and biofluids, resting-state fMRI (rs-fMRI) has been acquired in dedicated phases (notably ADNI-GO/ADNI-2 and later) at a subset of sites, with standardized instructions for task-free scanning (eyes-open) to promote uniform brain state [3]. Representative ADNI-2 rs-fMRI protocols (e.g., at Philips 3T sites) used a gradient-echo EPI sequence with TR/TE $\approx 3000/30$ ms, voxel size $\approx 3\times3\times3\,\mathrm{mm}^3$, $\sim$140 volumes ($\sim$7 min), and whole-brain coverage [94].

For this benchmark, we adopt a modern, reproducible preprocessing workflow (e.g., slice-timing and motion correction, susceptibility distortion handling where available, spatial normalization, nuisance regression, and careful quality control) implemented via `fMRIPrep` [30], which is robust to inter-site variability and designed to produce analysis-ready outputs for subsequent ROI-based time-series extraction and graph construction.

11.1.2 HCP S1200

The WU–Minn Human Connectome Project (HCP) S1200 release comprises >1,000 young adults (ages ∼22–35) with high-quality multimodal MRI and deep behavioral phenotyping [29]. Resting-state fMRI was acquired on a customized 3T Siemens Skyra using multiband EPI: TR = 0.72 s, TE = 33.1 ms, flip angle = 52°, 2.0 mm isotropic voxels, multiband factor 8, whole-brain coverage, four runs (∼15 min each; ∼1,200 volumes per run) with both phase-encoding polarities [46, 83]. Data were distributed with the HCP Minimal Preprocessing Pipelines (MPP) that include gradient unwarping, EPI distortion correction, motion correction, registration to structural surfaces/volumes, and ICA-FIX denoising; these pipelines are now widely adopted as a gold-standard reference [38, 47].

HCP's exceptional spatiotemporal resolution and standardized preprocessing enable high signal-to-noise ROI time series and provide a strong ceiling for evaluating graph construction methods in healthy, high-quality data [37, 83].

11.1.3 ADHD-200

The ADHD-200 Sample aggregates resting-state fMRI and structural MRI from 973 participants across eight international sites, spanning children, adolescents, and young adults, with clinical labels (ADHD subtypes, typically developing controls) and phenotypic measures [12, 67]. As a multi-site initiative, acquisition parameters vary by site (scanner vendor/model, TR typically in the ∼2–3 s range, 3–4 mm voxels, and run durations of roughly 5–8 min), a feature documented and summarized by the ADHD-200 Preprocessed repository [9]. To standardize access, three community preprocessing baselines were released: (i) *Athena* (AFNI+FSL) providing nuisance regression and band-pass filtered time series and derivatives, (ii) *Burner* (SPM) emphasizing voxel-based morphometry, and (iii) *NIAK* (PSOM) featuring ICA-based denoising and rigorous QC; these baselines remain useful reference points for ROI time-series extraction and graph building [9].

ADHD-200's multi-site heterogeneity and clinically meaningful labels make it a stringent evaluation context for robustness and generalization of ROI-based graph construction methods [9, 67].

11.2 Benchmark Framework

Figure 2 summarizes the end-to-end workflow from raw fMRI to graph-based prediction. BOLD images are first preprocessed to obtain ROI-level time series; these are parcellated into brain regions, converted into connectivity matrices either static or dynamic, assembled as graphs, and finally analyzed with a graph neural network to produce clinical predictions.

In the preprocessing stage, brain extraction removes non-brain tissues such as skull and scalp to reduce spurious signal and improve downstream alignment. Slice timing correction compensates for inter-slice acquisition delays so that time points are aligned across slices. Motion correction and realignment estimate head movement and resample volumes to a common reference, mitigating motion-induced variability. Co-registration aligns functional images to the subject's structural image, enabling anatomically informed region mapping. Normalization warps data to a common template space such as MNI to support group analyses and atlas-based region definitions. Smoothing applies a spatial kernel to improve signal-to-noise ratio and tolerance to residual misalignment. The output of this stage is an ROI time series representation suitable for connectivity analysis.

Parcellation maps the signals to representative atlases of complementary design and scale. AAL116 provides an anatomical definition with 116 regions of interest and is widely used for interpretability and comparison across studies. Schaefer100 defines 100 functionally coherent regions based on intrinsic connectivity gradients. Power264 provides 264 regions derived from data-driven clustering, offering finer resolution of functional loci. These parcellations allow the study of how different node definitions affect downstream graphs and predictions.

Connectivity construction computes relationships between ROI time series either as a static matrix aggregated across the entire run or as dynamic matrices from sliding windows that capture temporal fluctuations. Typical measures include Pearson correlation for linear association, partial correlation for direct associations, cosine similarity for directional similarity, wavelet coherence for frequency-specific coupling, cross-correlation for lagged interactions, and higher-order functional connectivity. Each method yields a weighted matrix that can be filtered or thresholded for subsequent analysis.

The final step is graph assembly and learning. Nodes from parcellation combined with connectivity measures form subject-level graphs. These graphs are then used as input to a graph neural network, consisting of stacked graph convolution layers followed by a multilayer perceptron, to perform training

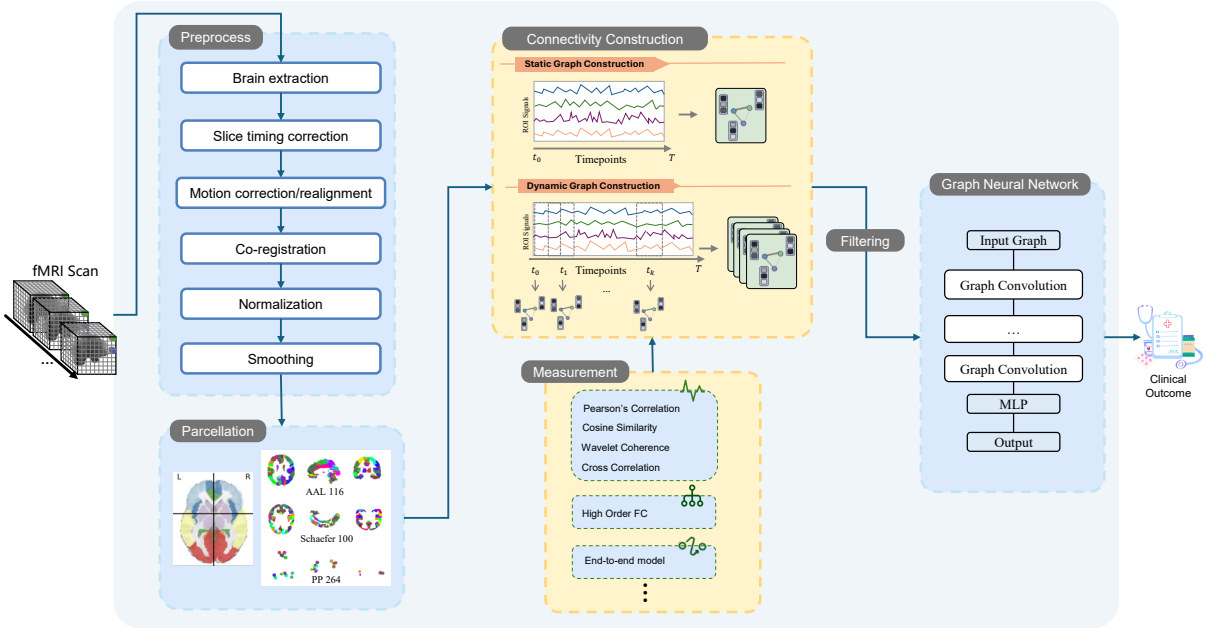

Figure 2: General workflow of the experiments. Raw fMRI data are preprocessed to generate ROI-level time series, parcellated into brain regions, transformed into connectivity matrices using either static or dynamic strategies, and assembled as graphs. These graphs are then analyzed by a graph neural network to produce clinical predictions.

and evaluation. The network outputs task-specific predictions and supports interpretation at the level of graph topology and node connectivity.

Table 3: Performance comparison of connectivity constructions for HC vs AD classification on the ADNI cohort using the **AAL116** parcellation and 30% Absolute Thresholding sparsification. Report mean ± s.d. over 5 folds.

Group	Connectivity	AUROC	Acc	Sens	Spec	F1	Prec
Static Pairwise	Pearson correlation	0.671±0.011	0.736±0.081	0.791±0.126	0.674±0.103	0.746±0.121	0.711±0.115
	Spearman correlation	0.664±0.088	0.750±0.074	0.790±0.110	**0.717±0.127**	**0.765±0.128**	0.740±0.106
	Kendall correlation	0.660±0.017	0.694±0.073	0.760±0.124	0.626±0.070	0.711±0.050	0.667±0.053
	Partial correlation	0.562±0.021	0.517±0.020	0.534±0.063	0.503±0.057	0.523±0.041	0.512±0.035
	Mutual information	0.609±0.049	0.590±0.074	0.598±0.041	0.583±0.116	0.592±0.110	0.592±0.126
	Distance correlation	**0.695±0.040**	0.698±0.079	0.786±0.084	0.600±0.123	0.733±0.065	0.675±0.096
	Coherence	0.592±0.049	0.553±0.008	0.512±0.104	0.593±0.097	0.533±0.097	0.557±0.115
	Cross-correlation	0.552±0.008	0.536±0.075	0.538±0.116	0.540±0.047	0.541±0.074	0.536±0.075
	Wavelet coherence	0.560±0.024	0.469±0.008	0.437±0.115	0.507±0.092	0.454±0.116	0.470±0.123
	Granger causality	0.578±0.018	0.659±0.067	0.669±0.128	0.654±0.127	0.663±0.099	0.657±0.095
	Cosine similarity	0.690±0.061	0.675±0.066	0.765±0.118	0.577±0.103	0.702±0.071	0.650±0.200
	Euclidean distance	0.665±0.054	**0.751±0.067**	**0.860±0.112**	0.651±0.093	0.771±0.185	0.702±0.095
	Patel kappa	0.601±0.048	0.519±0.088	0.521±0.066	0.520±0.121	0.518±0.098	0.534±0.111
	Patel tau	0.603±0.044	0.634±0.080	0.589±0.091	0.669±0.105	0.614±0.097	0.638±0.159
	LiNGAM	0.551±0.017	0.528±0.080	0.498±0.126	0.562±0.090	0.513±0.125	0.526±0.099
	PLV	0.555±0.016	0.541±0.053	0.560±0.046	0.534±0.102	0.554±0.052	0.555±0.055
	PSI	0.594±0.085	0.580±0.087	0.587±0.057	0.573±0.130	0.582±0.063	0.586±0.093
Static HOFC	Correlation's correlation	0.689±0.011	0.734±0.080	0.804±0.117	0.650±0.129	0.744±0.130	0.698±0.125
	tHOFC	0.608±0.050	0.572±0.065	0.537±0.063	0.598±0.168	0.557±0.072	0.578±0.074
	aHOFC	0.601±0.052	0.587±0.077	0.601±0.094	0.573±0.117	0.577±0.129	0.573±0.105
	dHOFC	0.552±0.017	0.541±0.072	0.509±0.099	0.570±0.113	0.520±0.096	0.532±0.092
Static End-to-end	FBNetGen	0.666±0.072	0.734±0.084	0.794±0.091	0.687±0.125	0.763±0.125	**0.745±0.107**
	DABNet	0.565±0.022	0.500±0.069	0.490±0.114	0.510±0.100	0.482±0.076	0.477±0.090
	Graph autoencoder	0.566±0.023	0.617±0.069	0.601±0.099	0.636±0.160	0.616±0.123	0.627±0.102

Table 4: Performance comparison of connectivity constructions for gender classification on the **HCP** cohort using the AAL116 parcellation and 30% Absolute Thresholding sparsification. Report mean ± s.d. over 5 folds.

Group	**Connectivity**	**AUROC**	**Acc**	**Sens**	**Spec**	**F1**	**Prec**
Static Pairwise	Pearson correlation	**0.844±0.047**	0.782±0.076	0.875±0.094	0.713±0.107	0.809±0.104	**0.759±0.123**
	Spearman correlation	0.735±0.051	0.767±0.087	0.850±0.090	0.688±0.096	0.788±0.105	0.736±0.098
	Kendall correlation	0.741±0.059	0.659±0.082	0.742±0.110	0.580±0.121	0.689±0.116	0.642±0.122
	Partial correlation	0.680±0.020	0.566±0.070	0.524±0.109	0.606±0.086	0.540±0.063	0.570±0.102
	Mutual information	0.702±0.022	0.601±0.009	0.622±0.073	0.576±0.069	0.608±0.192	0.593±0.061
	Distance correlation	0.696±0.031	0.502±0.086	0.639±0.131	0.375±0.117	0.570±0.099	0.504±0.128
	Coherence	0.697±0.020	0.677±0.070	0.675±0.109	0.691±0.104	0.684±0.108	0.701±0.113
	Cross-correlation	0.680±0.016	0.535±0.065	0.541±0.115	0.530±0.103	0.539±0.099	0.538±0.079
	Wavelet coherence	0.691±0.020	0.561±0.011	0.540±0.131	0.586±0.090	0.556±0.110	0.573±0.126
	Granger causality	0.680±0.021	0.575±0.051	0.533±0.058	0.599±0.093	0.553±0.041	0.578±0.034
	Cosine similarity	0.794±0.055	**0.804±0.068**	**0.894±0.131**	0.708±0.091	0.814±0.096	0.743±0.093
	Euclidean distance	0.734±0.053	0.668±0.077	0.730±0.122	0.603±0.086	0.687±0.094	0.651±0.097
	Patel kappa	0.693±0.011	0.596±0.081	0.625±0.099	0.566±0.161	0.608±0.113	0.595±0.120
	Patel tau	0.764±0.046	0.749±0.052	0.777±0.094	0.727±0.053	0.751±0.050	0.735±0.060
	LiNGAM	0.721±0.057	0.668±0.066	0.684±0.117	0.648±0.063	0.665±0.122	0.653±0.127
	PLV	0.680±0.018	0.499±0.029	0.488±0.094	0.502±0.044	0.493±0.050	0.501±0.043
	PSI	0.680±0.020	0.529±0.033	0.597±0.066	0.469±0.051	0.566±0.040	0.530±0.034
Static HOFC	Correlation's correlation	0.752±0.036	0.798±0.089	0.868±0.091	**0.729±0.105**	**0.821±0.095**	0.770±0.124
	tHOFC	0.701±0.018	0.647±0.085	0.669±0.098	0.617±0.101	0.651±0.095	0.638±0.115
	aHOFC	0.682±0.020	0.559±0.012	0.599±0.046	0.515±0.078	0.582±0.036	0.560±0.058
	dHOFC	0.680±0.019	0.526±0.010	0.570±0.129	0.500±0.112	0.549±0.114	0.529±0.104
Static End-to-end	FBNetGen	0.738±0.078	0.658±0.079	0.620±0.111	0.706±0.120	0.643±0.106	0.679±0.122
	DABNet	0.722±0.048	0.634±0.073	0.666±0.096	0.607±0.113	0.649±0.072	0.635±0.111
	Graph autoencoder	0.707±0.054	0.561±0.079	0.554±0.095	0.553±0.107	0.559±0.125	0.557±0.101

Table 5: Performance comparison of connectivity constructions for HC vs ADHD classification on the **ADHD** cohort using the AAL116 parcellation and **30% Absolute Thresholding** sparsification. Report mean ± s.d. over 5 folds.

Group	Connectivity	AUROC	Acc	Sens	Spec	F1	Prec
Static Pairwise	Pearson correlation	0.611±0.075	0.620±0.066	0.685±0.103	0.572±0.094	0.655±0.120	0.630±0.110
	Spearman correlation	0.639±0.045	0.633±0.053	0.743±0.174	0.514±0.041	0.674±0.072	0.615±0.037
	Kendall correlation	0.662±0.044	0.696±0.078	0.754±0.170	0.643±0.191	0.709±0.078	**0.673±0.080**
	Partial correlation	0.570±0.081	0.577±0.075	0.595±0.120	0.557±0.073	0.590±0.095	0.573±0.090
	Mutual information	0.578±0.065	0.471±0.069	0.450±0.044	0.494±0.102	0.460±0.085	0.481±0.104
	Distance correlation	0.608±0.059	0.703±0.069	**0.781±0.089**	0.624±0.130	**0.723±0.085**	0.680±0.099
	Coherence	0.538±0.021	0.532±0.085	0.547±0.092	0.512±0.112	0.556±0.102	0.564±0.099
	Cross-correlation	0.553±0.051	0.504±0.050	0.504±0.174	0.497±0.050	0.507±0.079	0.509±0.064
	Wavelet coherence	0.592±0.066	0.568±0.082	0.506±0.093	0.629±0.087	0.544±0.071	0.583±0.067
	Granger causality	0.550±0.010	0.553±0.074	0.558±0.119	0.543±0.055	0.540±0.062	0.538±0.042
	Cosine similarity	0.631±0.070	0.613±0.080	0.736±0.116	0.487±0.108	0.644±0.112	0.576±0.098
	Euclidean distance	**0.692±0.045**	**0.734±0.075**	0.811±0.099	0.667±0.121	0.765±0.088	0.716±0.104
	Patel kappa	0.619±0.070	0.588±0.055	0.536±0.072	0.647±0.030	0.567±0.059	0.598±0.060
	Patel tau	0.585±0.064	0.550±0.079	0.526±0.072	0.575±0.102	0.532±0.076	0.542±0.104
	LiNGAM	0.561±0.051	0.517±0.028	0.458±0.035	0.584±0.056	0.490±0.031	0.528±0.044
	PLV	0.550±0.052	0.543±0.076	0.501±0.088	0.570±0.094	0.527±0.054	0.545±0.068
	PSI	0.542±0.009	0.503±0.081	0.506±0.103	0.518±0.117	0.507±0.118	0.520±0.126
Static HOFC	Correlation's correlation	0.612±0.043	0.609±0.084	0.722±0.126	0.492±0.113	0.643±0.110	0.590±0.113
	tHOFC	0.554±0.028	0.484±0.056	0.505±0.041	0.465±0.084	0.490±0.020	0.485±0.035
	aHOFC	0.572±0.030	0.637±0.085	0.630±0.068	**0.649±0.117**	0.635±0.117	0.650±0.111
	dHOFC	0.566±0.055	0.463±0.082	0.514±0.058	0.400±0.104	0.486±0.111	0.463±0.082
Static End-to-end	FBNetGen	0.611±0.021	0.672±0.040	0.731±0.183	0.604±0.051	0.686±0.079	0.651±0.038
	DABNet	0.540±0.020	0.533±0.061	0.493±0.052	0.585±0.123	0.506±0.188	0.539±0.093
	Graph autoencoder	0.536±0.018	0.477±0.062	0.446±0.076	0.513±0.099	0.459±0.093	0.477±0.068

NeuroGraph

In this study, we reproduced the GNN model from the neurograph library. The model is built using a series of graph convolutional layers, specifically designed to handle node-level features, and incorporates residual connections and batch normalization for improved training efficiency and model generalization. Given an input graph $G = (V, E)$ with node features $X \in \mathbb{R}^{n \times f}$ and edge indices $E \in \mathbb{R}^{2 \times m}$, the forward pass of the model can be described as follows:

Graph Convolution: At each layer, the input features X are transformed using graph convolution operations, denoted as $\text{Conv}_i(X, E)$. The transformation is non-linear, and the output is passed through a tanh activation function:

$$X_{i+1} = \tanh(\text{Conv}_i(X, E))$$

Residual Connections The output of each layer is combined with the input via residual connections. This helps in training deeper networks by mitigating vanishing gradients:

$$X_{\text{res}} = X_i + X_{i+1}$$

Feature Aggregation The node-level features from each graph are aggregated using the Mean Aggregation approach:

$$h_{\text{agg}} = \frac{1}{n} \sum_{i=1}^{n} X_{\text{res}}(i)$$

Classification The aggregated features h_{agg} are passed through the MLP for final classification. The MLP consists of several layers with batch normalization, ReLU activations, and dropout for regularization:

$$\hat{y} = \text{MLP}(h_{\text{agg}})$$

Output The final output is a set of class probabilities produced using the softmax function:

$$\hat{y} = \text{softmax}(x)$$

The model is trained using standard gradient-based optimization techniques, with the loss function typically being the cross-entropy loss for classification tasks. The model is designed to run efficiently

on both CPUs and GPUs, leveraging PyTorch's GPU acceleration for training large models on graph-structured data.

11.3 Results

Here we address the questions outlined earlier by conducting a series of experiments, including the evaluation of different atlases node configurations, the implications of the filtering scheme in brain graphs, and mainly, the connectivity pattern. Each experiment aligns with a question, thereby paving the way for comprehensive analysis and definitive conclusions.

11.4 Evaluation

11.4.1 Causal Analysis

11.4.2

11.5 Open-Source Resources

To promote transparency and reproducibility, we provide an open-source Python package, `brainnet-graph`, available via PyPI.[1] This lightweight toolkit supports multiple correlation- and distance-based methods for constructing functional brain graphs from ROI-level fMRI BOLD signals, with outputs in either PyTorch Geometric (`.pt`) or edge list (`.csv`) formats. It includes both a Python API and a command-line interface for seamless integration into neuroimaging pipelines.

Comprehensive documentation, tutorials, and usage examples are hosted online.[2] These resources are intended to ensure accessibility for researchers in both neuroscience and machine learning, facilitating adoption and extension of the benchmark framework.

11.6 Intelligent Agent

While open-source code ensures transparency and reproducibility, additional support is often required for researchers who are unfamiliar with the technical details or who seek rapid prototyping. To this end, we introduce an intelligent agent designed to facilitate interactive use of the framework. The agent integrates project-specific knowledge—including code, documentation, and experimental records—and employs retrieval-augmented generation (RAG) to answer user queries with contextually relevant information (see Fig. 1).

Beyond passive assistance, the agent is capable of executing experiments on demand, returning results in standardized formats, and providing explanatory feedback to aid interpretation. In this way, it serves as both a computational assistant and an educational interface, lowering the barrier to entry for non-specialists while also supporting advanced users in efficiently deploying and analyzing complex brain network pipelines.

More broadly, the agent embodies a shift toward human-centered computational neuroscience. By providing natural language access to sophisticated pipelines, it bridges the gap between technical infrastructure and scientific inquiry. This design empowers users to focus on formulating neuroscientific questions rather than struggling with implementation details, thus accelerating discovery. Moreover, as the agent continuously learns from both structured resources and user interactions, it has the potential to evolve into a dynamic research companion—capable not only of reproducing results, but also of guiding methodological choices and highlighting best practices. In this sense, the agent is not merely a deployment tool, but a catalyst for more accessible, reproducible, and collaborative connectomics research.

12 Discussion

12.1 Summary of Benchmark Findings

We introduced *BrainNet*, a systematic benchmark for ROI-based functional brain network construction in resting-state fMRI. Across node definitions, connectivity measures, and sparsification strategies, we observed that the *graph construction pipeline* is a primary determinant of downstream behavior. Pipeline

[1] `https://pypi.org/project/brainnet-graph/`
[2] `https://brainnet-graph.readthedocs.io/`

choices influenced reliability (e.g., test–retest consistency and motion robustness), sensitivity to inter-subject differences, and classification performance under a unified training/validation protocol. Although variability was substantial, a compact subset of pipelines satisfied all evaluation criteria across datasets, providing practical defaults for reproducible analyses.

12.2 Variability Across Graph Construction Pipelines

Node definition. Parcellation scale and principle (anatomical, functional, or data-driven) materially affected network topology and variance. Finer parcellations offered greater detail but increased susceptibility to noise and alignment errors; very coarse atlases improved stability but risked missing meso-scale organization. Subject-specific parcellations improved subject-level fit in some settings but complicated cross-subject comparability. In aggregate, moderate granularity provided a reasonable balance between stability and representational capacity.

Connectivity measures. Edge definitions shaped graph structure and downstream signal. Baseline Pearson correlation produced workable static graphs, whereas multivariate estimators (e.g., partial correlation/precision) often improved reliability by attenuating indirect associations. Higher-order functional connectivity (HOFC) complemented low-order edges by capturing similarity of connectivity profiles. Dynamic estimates helped when scan duration and data quality supported time-varying modeling, but added variance when those conditions were not met. Learned-edge approaches improved performance in several settings, provided that appropriate regularization and validation prevented overfitting to nuisance variation.

Sparsification and thresholding. Filtering decisions (absolute vs. proportional thresholds, graph-theoretic backbones, and related schemes) had outsized effects on density, topology, and between-subject comparability. Absolute cutoffs produced variable densities across subjects; proportional schemes standardized density but could discard weak yet meaningful edges in low-SNR data. Performance rankings were sensitive to density, indicating that sparsification is a first-class hyperparameter rather than a minor post-processing step.

12.3 Robust Configurations Across Datasets

Despite heterogeneity across ADNI, HCP, and ADHD-200, certain configurations were consistently robust under our criteria. Pipelines coupling (i) moderate-resolution parcellations, (ii) multivariate connectivity estimators (e.g., regularized partial correlation), and (iii) proportional sparsification at appropriate densities tended to yield graphs with favorable trade-offs: higher test–retest stability, reduced motion sensitivity, and preserved discriminability. These configurations provide practical defaults and a strong baseline for subsequent methodological exploration.

12.4 Methodological Pitfalls and Reproducibility

Two classes of choices were repeatedly associated with poorer reproducibility or sensitivity. First, aggressive or idiosyncratic thresholding (e.g., high absolute cutoffs or unqualified binarization) produced density-driven artifacts and unstable graph metrics. Treating density as a tunable parameter and reporting results across a sensible range mitigated this risk. Second, unconstrained learned-edge models sometimes overfit site- or scanner-specific idiosyncrasies. Domain-informed constraints (symmetry, sparsity, positive definiteness) and site-aware, nested validation reduced this failure mode.

12.5 Implications for Downstream Machine Learning

The quality of the connectome meaningfully limits downstream models. In our setting, improving the *graph* (e.g., via multivariate connectivity and principled sparsification) often yielded larger gains than altering the classifier, including GNN variants, when the classifier and training protocol were otherwise held fixed. We recommend elevating graph construction to a core model-selection dimension: evaluate multiple node–edge–threshold combinations alongside standard ML hyperparameters, and report variability to guard against construction-specific artifacts.

12.6 Reliability and Generalizability

Pipelines with rigorous nuisance control (e.g., motion regression and scrubbing) and density-aware filtering produced more stable networks. Test–retest evaluation is informative for short-timescale stability when no true change is expected. For cross-dataset generalization, site-stratified splits and nested model selection reduced optimistic bias, and replications across cohorts clarified which pipelines transfer beyond a single sample. Reporting uncertainty (e.g., confidence intervals or variability bands across densities and atlases) further strengthens comparability.

12.7 Limitations

Our benchmark focuses on ROI-level graphs and a curated set of static and dynamic connectivity estimators under a unified preprocessing and training protocol. Voxel-/surface-level graphs, alternative preprocessing choices, causal/effective connectivity, and higher-order graph formalisms (e.g., hypergraphs, multiplex networks) were not exhaustively evaluated. We fixed the downstream classifier family to isolate construction effects; different learners may interact differently with specific graph inputs. Cohort heterogeneity (age, health status, acquisition) is both a strength for generalization tests and a constraint on universal recommendations.

12.8 Future Directions

Promising avenues include: (i) multi-resolution and individualized parcellations with principled cross-subject harmonization; (ii) data-efficient dynamic connectivity modeling beyond sliding windows; (iii) integration of low-/high-order, static/dynamic, and structural/functional information within unified graph formalisms; (iv) regularized learned-edge methods with uncertainty quantification and interpretability; and (v) community standards for reporting atlas, connectivity, sparsification, and evaluation protocols. Extending *BrainNet* along these axes can advance reproducibility and utility for both neuroscientific inference and clinical applications.

13 Conclusion

We introduced *BrainNet*, a systematic benchmark and taxonomy for functional brain network construction that separates the contribution of graph generation from downstream modeling. Across three public cohorts and a fixed GNN classifier, we showed that: (i) pipeline choice—node, edge, and sparsification—dominates predictive variance; (ii) nontrivial edge models (partial/precision, HOFC, learned edges) frequently surpass Pearson baselines; (iii) sparsity controls can invert method rankings; and (iv) dynamic graphs provide additional gains when adequately supported by data and validation. We distilled these findings into a decision map and practical defaults to help researchers select and tune pipelines aligned with their scientific goals.

By releasing a transparent evaluation protocol and resources, we aim to elevate graph construction to a first-class design variable in connectomics, enabling robust, comparable biomarkers and accelerating reproducible science. Future work will extend BrainNet to individualized and multi-scale nodes, richer dynamic and causal models, and tighter integration with structural connectivity and physiology, moving the field toward principled, end-to-end neurobiologically grounded graph generation.

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